

coursera

CGS402

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ASSIGNMENT - 2

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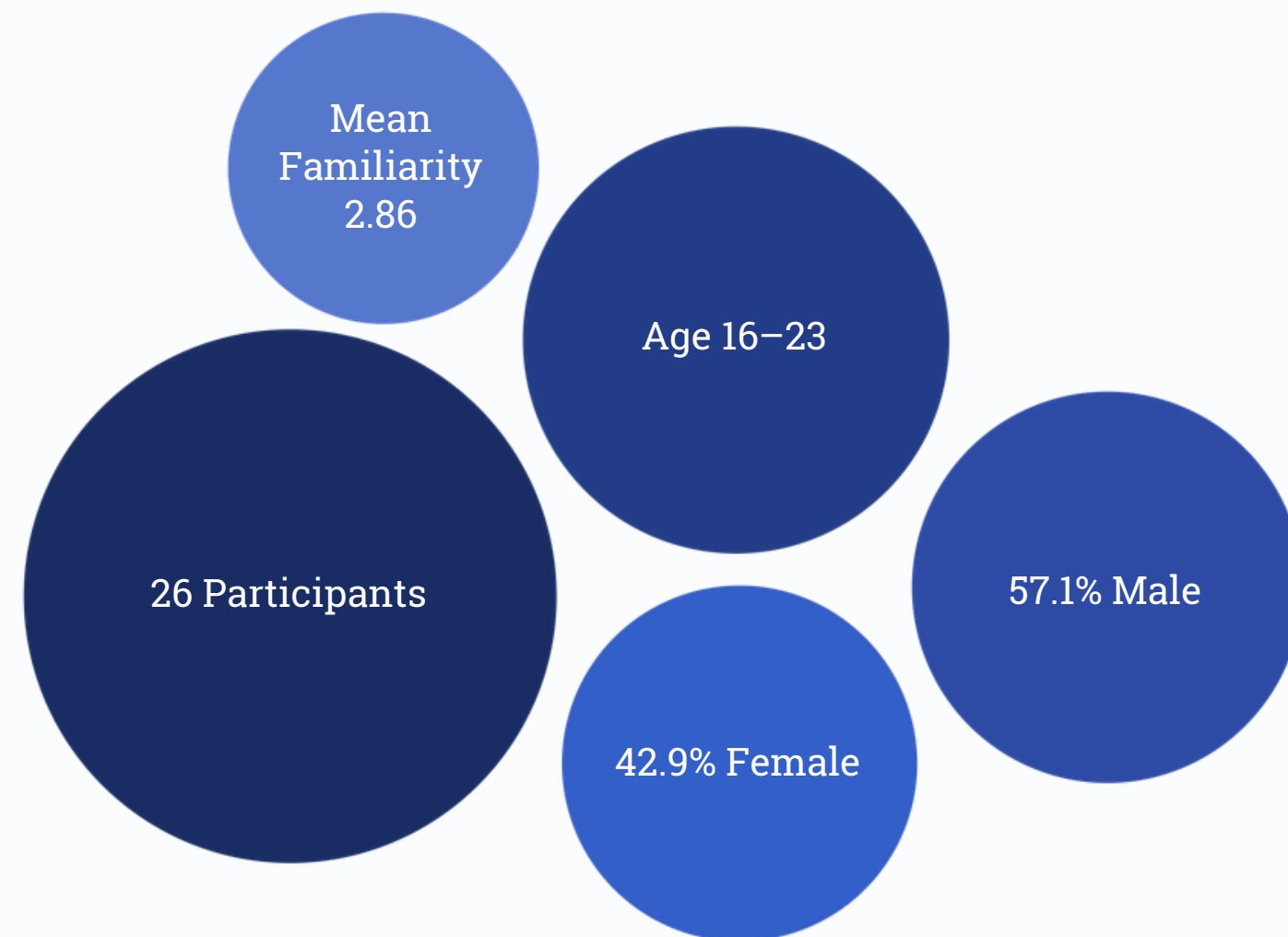
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Coursera: Platform Overview

Founded in 2012, Coursera is a leading online learning platform offering courses, professional certificates, and degree programs from top universities worldwide — spanning science & technology, business, and the humanities through video lectures, interactive assignments, and peer-reviewed assessments.

Selected for its widespread campus use, multi-disciplinary course range, and integration of cognitive science principles — structured content delivery, active recall through quizzes, and progressive skill building.





Memory

How is new material presented in the website?

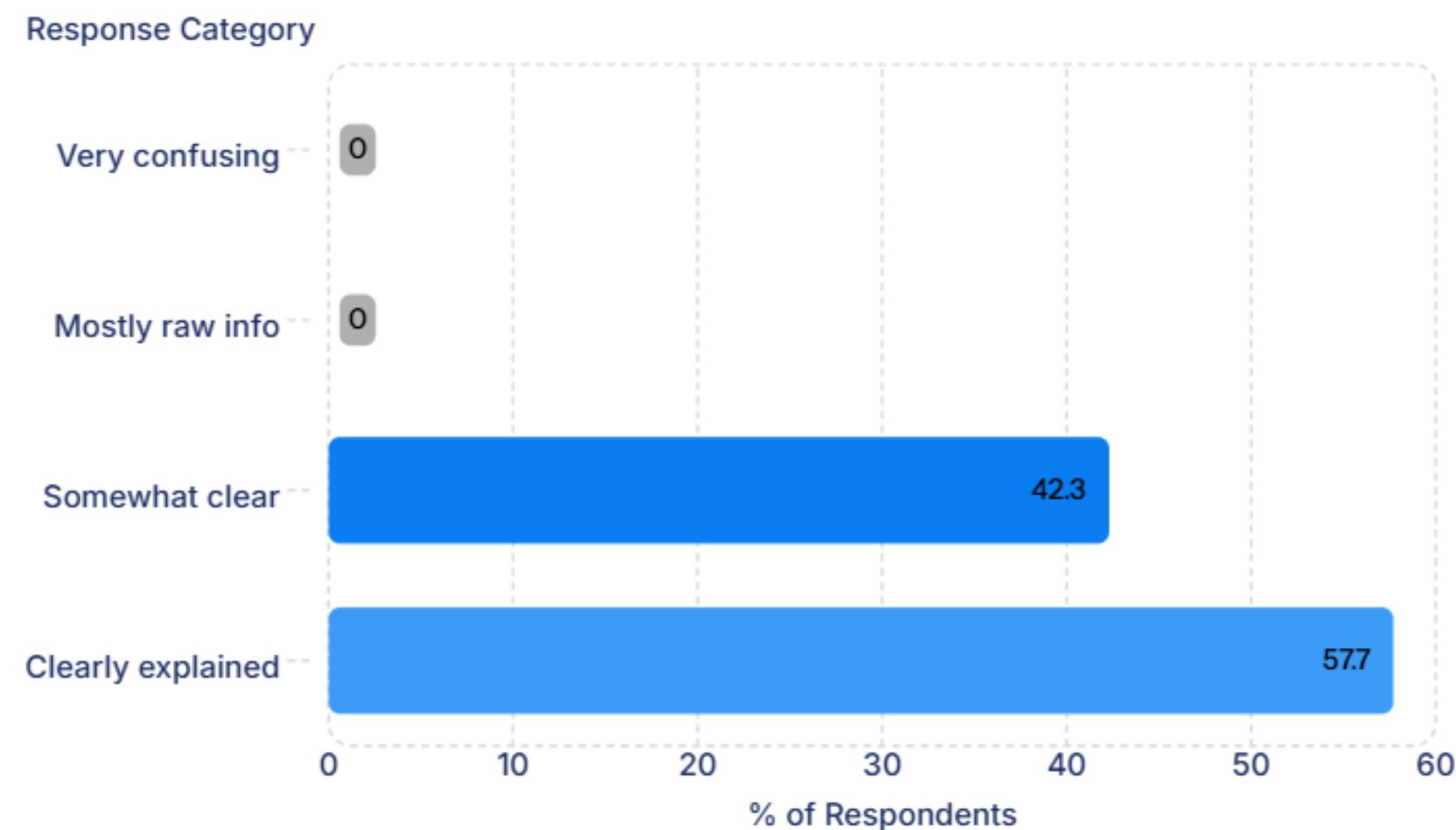


Figure 1: Q1 – Presentation of New Material. Categorical responses (n = 26).

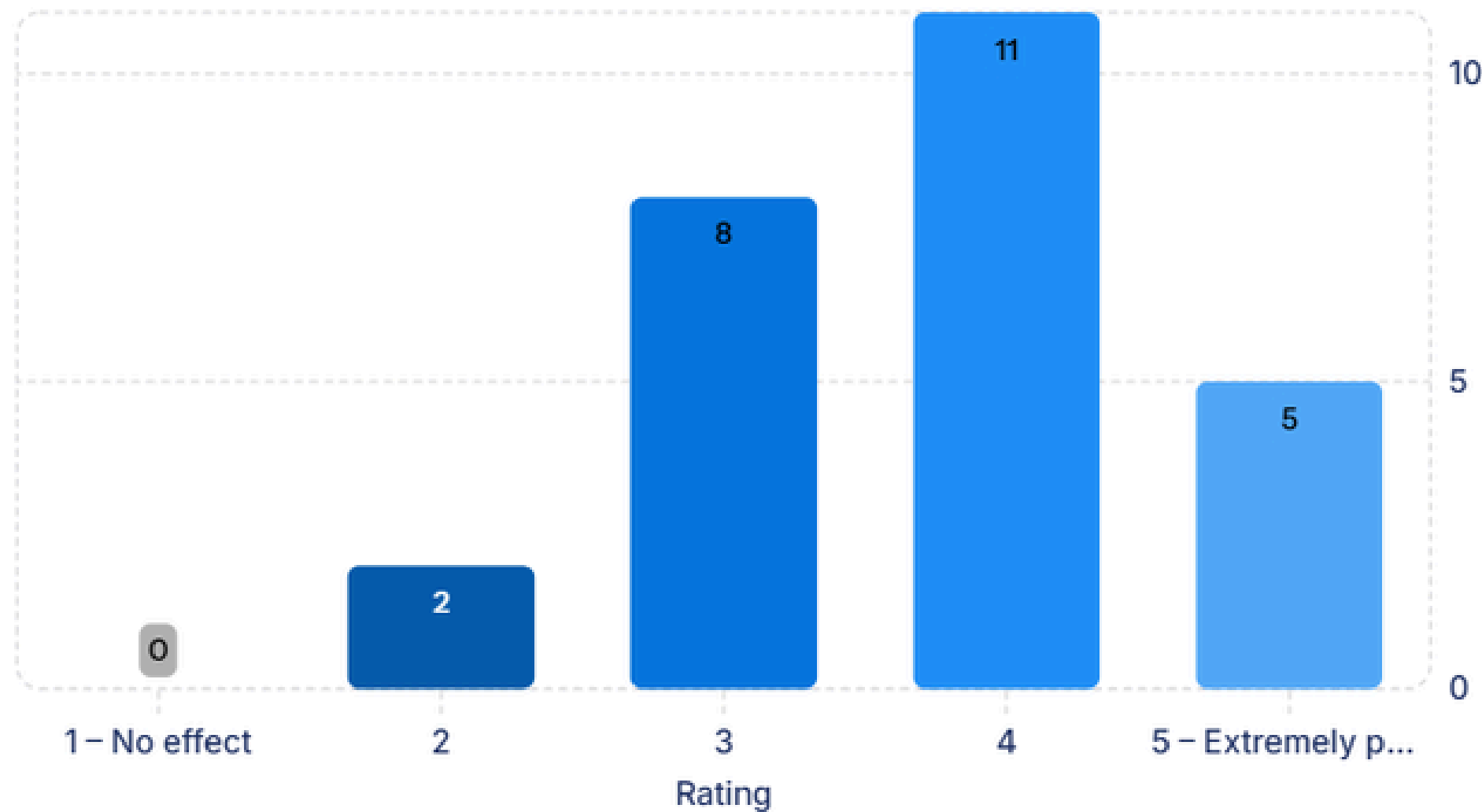
Key Finding

A majority (57.7%) found new material clearly explained with examples, visuals, and conceptual connections. The remaining 42.3% found the explanation only partial. No respondent reported raw or confusing presentation.

Dual Encoding

Coursera's use of verbal explanations along with visuals (dual encoding) likely explains why most users rated content as "clearly explained" (57.7%). The remaining 42.3% suggests that clarity also depends on the difficulty of the course. The absence of any "not clear" responses further indicates that dual encoding is generally effective.

How do game elements affect your focus?



"How do game elements (points, levels, rewards) affect your focus?"

Key Finding

Game elements are generally effective (mean = 3.73), with most ratings at 3 and 4, and no responses indicating they are completely ineffective.

Processing Efficiency Theory

According to Processing Efficiency Theory, moderate motivational cues improve attention and task focus without overloading cognitive resources. Game elements like points and streaks provide this optimal level of stimulation, helping users stay engaged. However, excessive reliance on such rewards may reduce intrinsic motivation.

Does the app revisits the previously learned material?

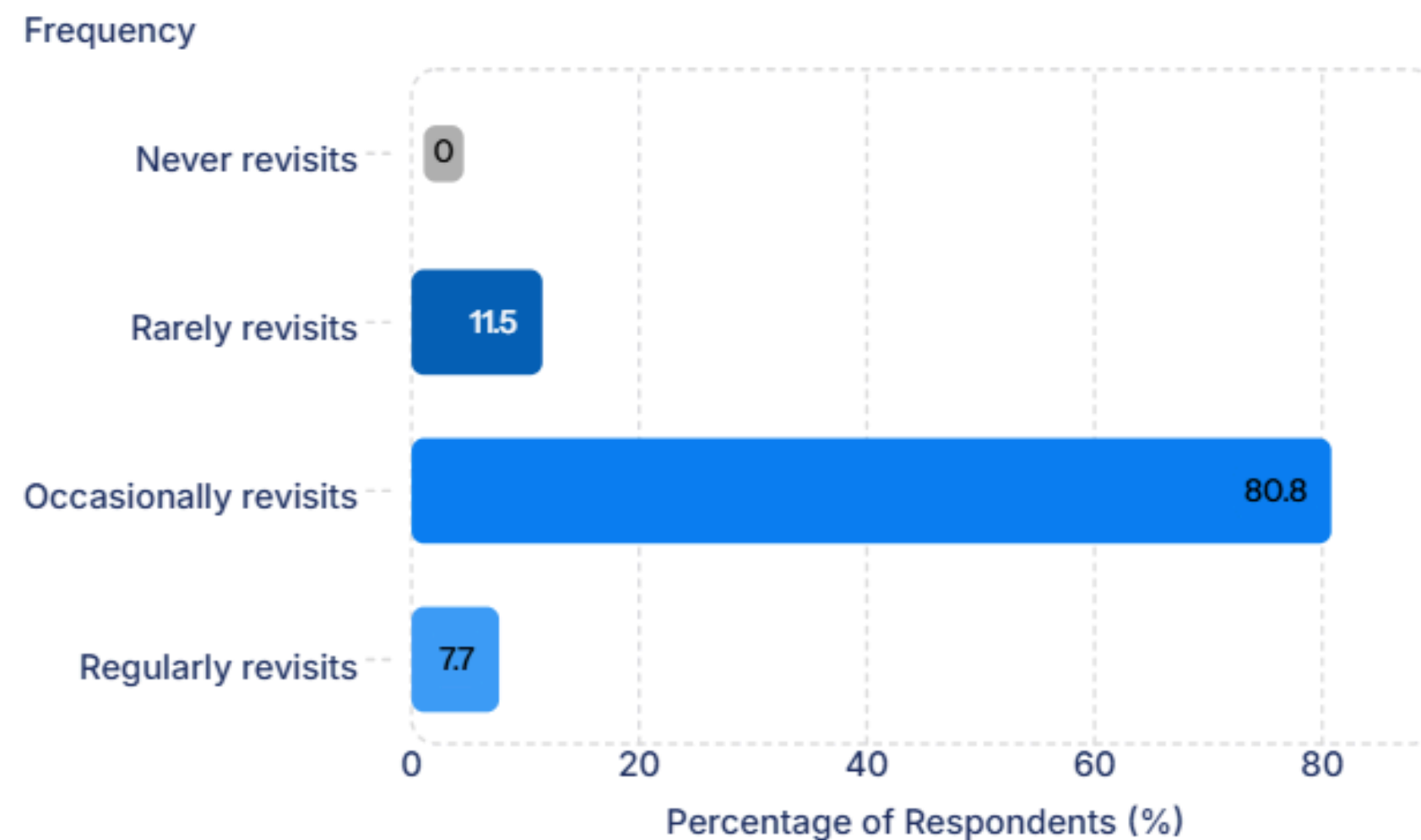


Figure 3: Q3 – Frequency of spaced content revisitation (n = 26).

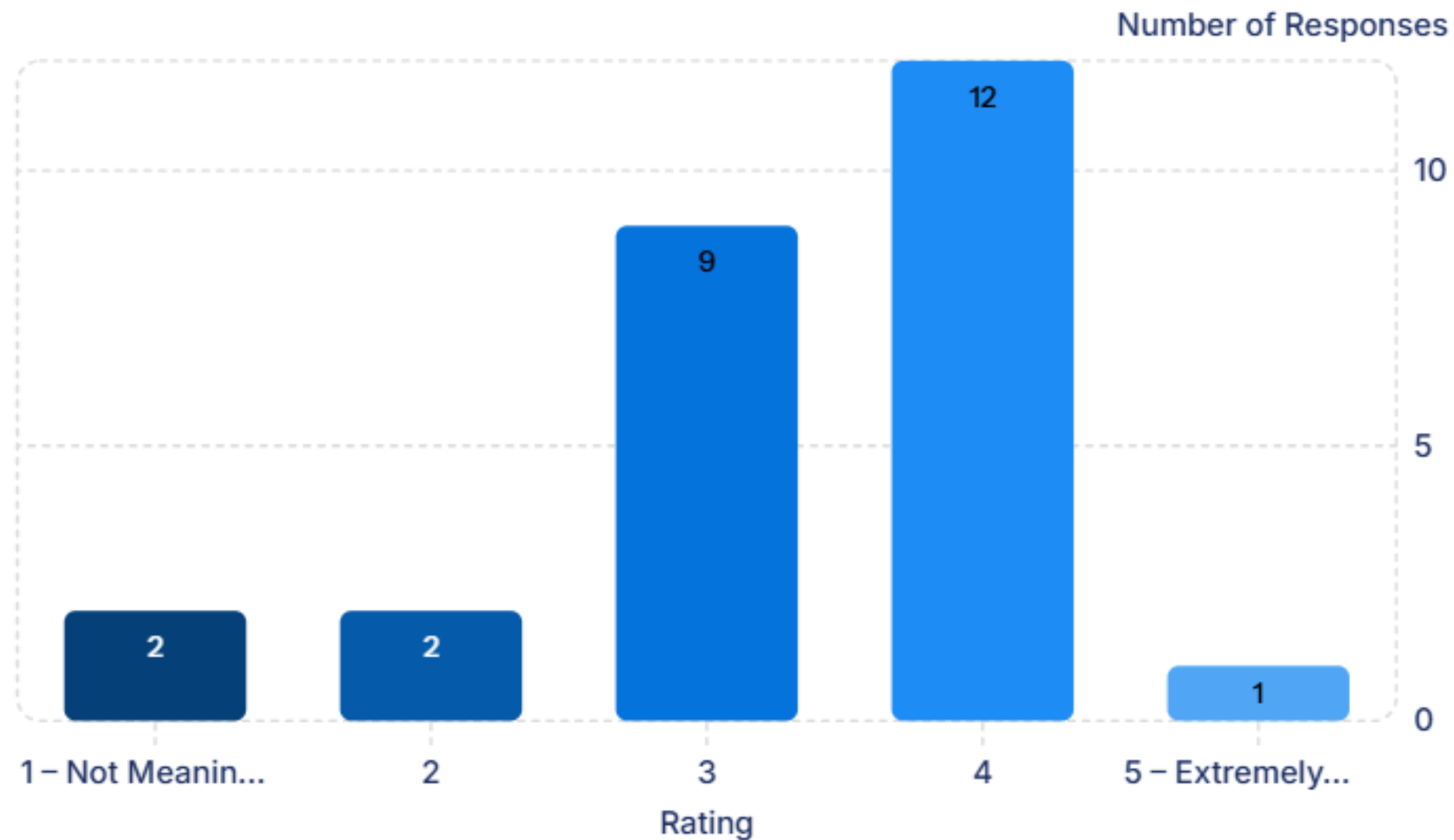
Key Finding

Most users occasionally revisit content (80.8%), while 11.5% rarely revisit and only 7.7% do so regularly, indicating a lack of consistent revision.

Spacing Effect

According to the Spacing Effect and Ebbinghaus' Forgetting Curve, memory fades rapidly without repeated review, while spaced revision strengthens retention. The dominance of "occasionally" suggests that revisiting is unstructured and inconsistent, reducing its effectiveness. Implementing systematic spaced repetition can improve long-term retention.

How meaningful is the content review?



Mean = 3.31 / 5 | Mode = 4 (46.2%) | n = 26

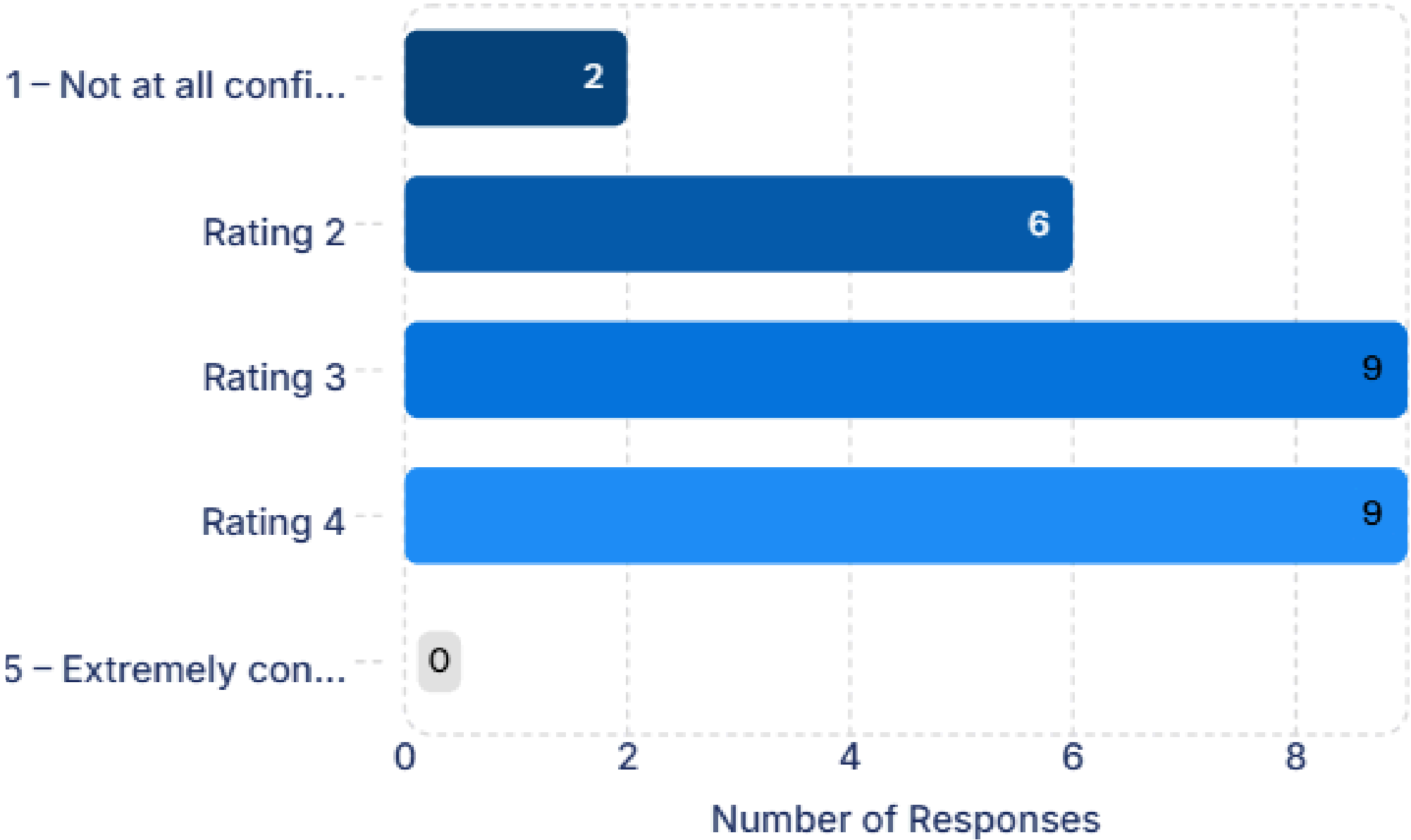
Key Finding

Most users rate review as moderately meaningful (mode = 4, 46.2%), but low ratings at 1 and 2 (7.7% each) bring the mean down to 3.31, with very few finding it extremely meaningful.

Elaborative Retrieval

Revisiting is most effective when it involves active recall rather than passive re-exposure. The moderate mean suggests that current reviews rely more on rereading or summaries, which have limited impact on retention. Incorporating elaborative retrieval methods, such as generative quizzes, can significantly improve learning outcomes.

How confident are you in retaining the content after 2 weeks?



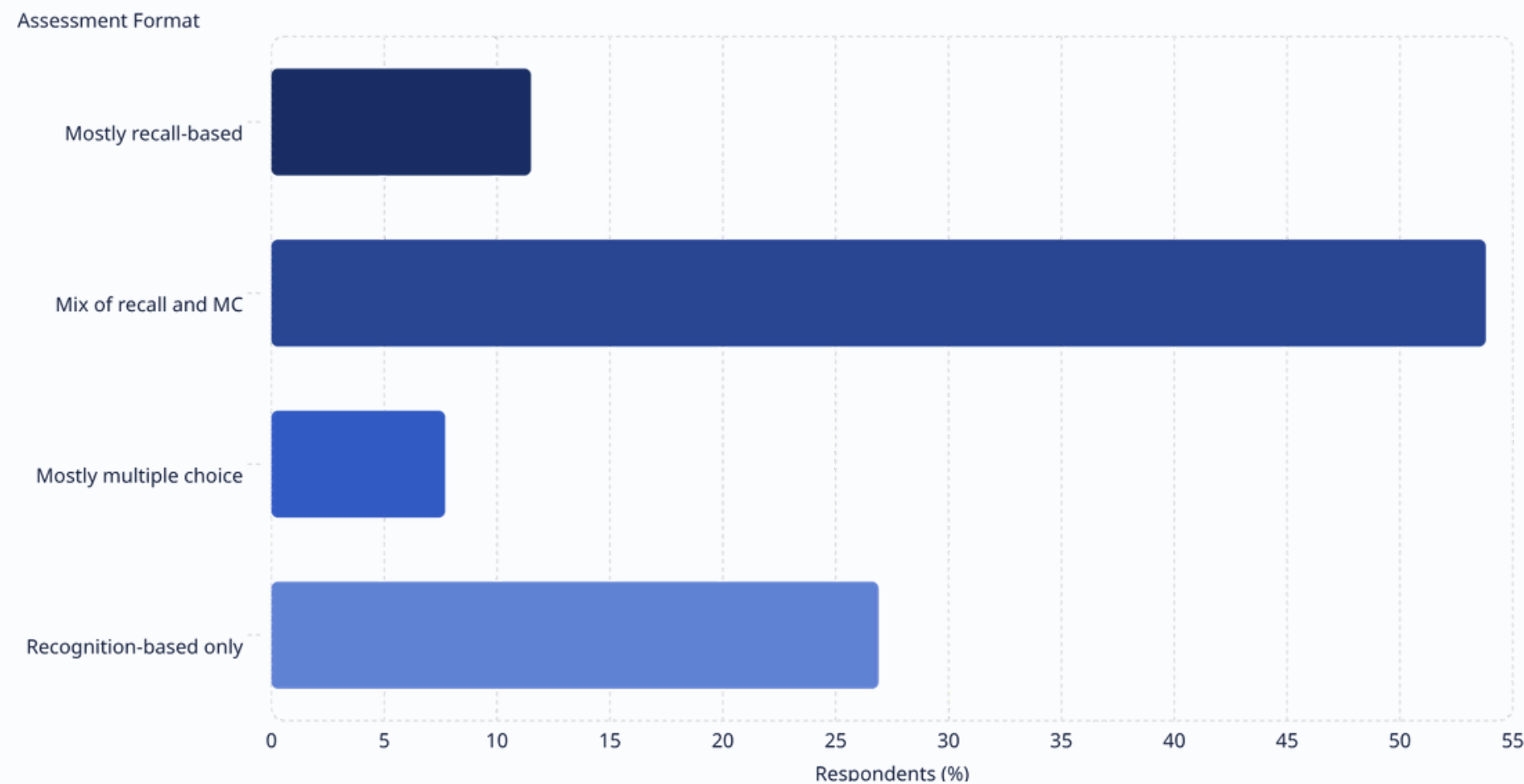
Key Finding

Confidence in long-term retention is low (mean = 2.96), the lowest across all questions. No respondent rated 5, with most responses clustered at 3 (34.6%) and 4 (34.6%), indicating only partial confidence.

DECAY WITH DISUSE

According to decay with disuse, memory fades without repeated retrieval. This is reinforced by earlier findings revisiting is mostly occasional rather than regular, and review quality is only moderately effective due to reliance on passive methods. This lack of consistent spaced repetition and active recall leads to weak consolidation and rapid forgetting.

How Does the App Test Understanding?



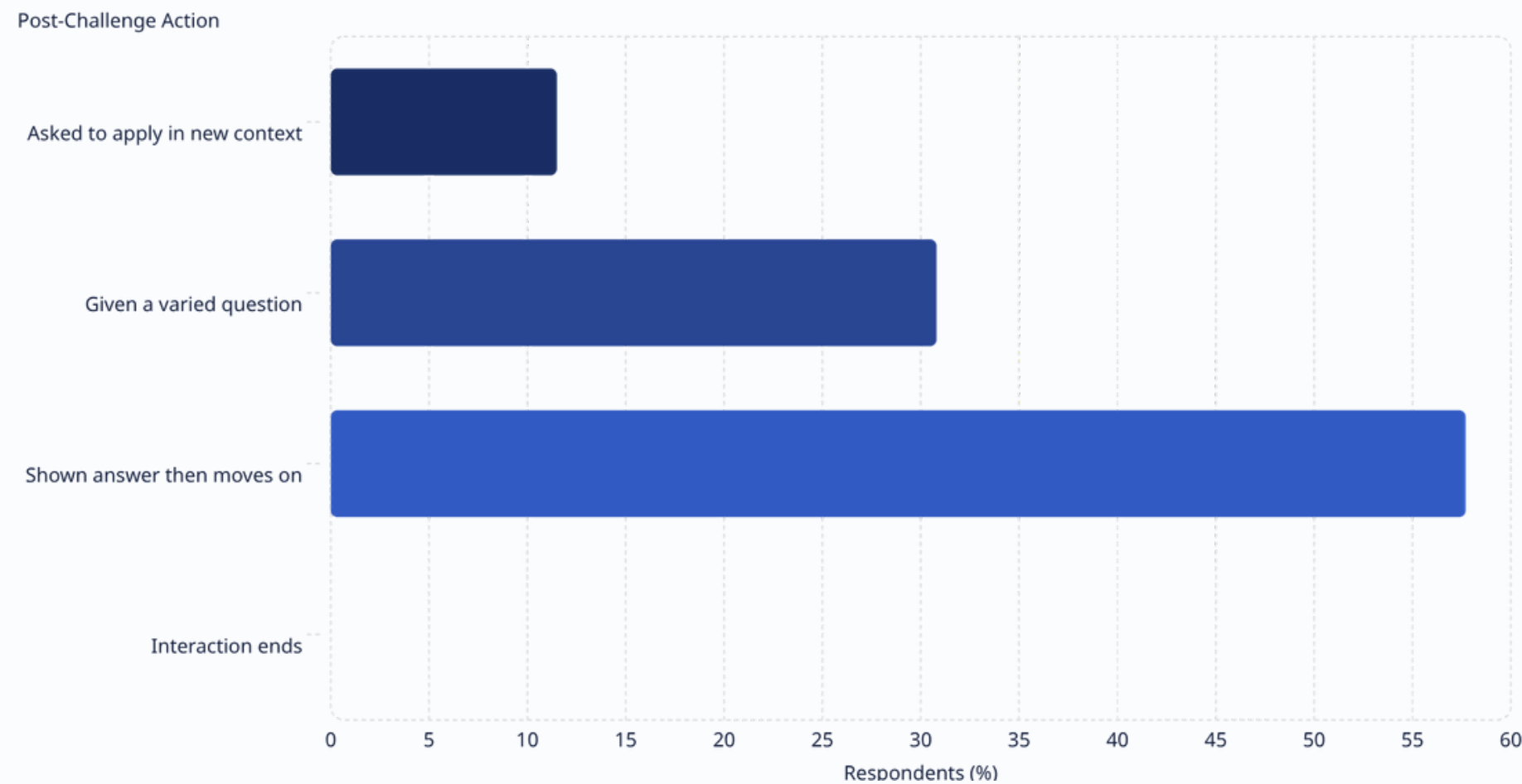
Key Finding

Over half (53.8%) perceived a balanced mix of recall and multiple-choice formats. Notably, **26.9% characterised assessments as purely recognition-based**, while only 11.5% reported primarily free-recall formats.

The Testing Effect

Free-recall testing produces significantly better retention than recognition tests. Recognition-based questions reduce retrieval difficulty and **minimize the Testing Effect**, making the 26.9% figure concerning.

What Happens After a Challenge?



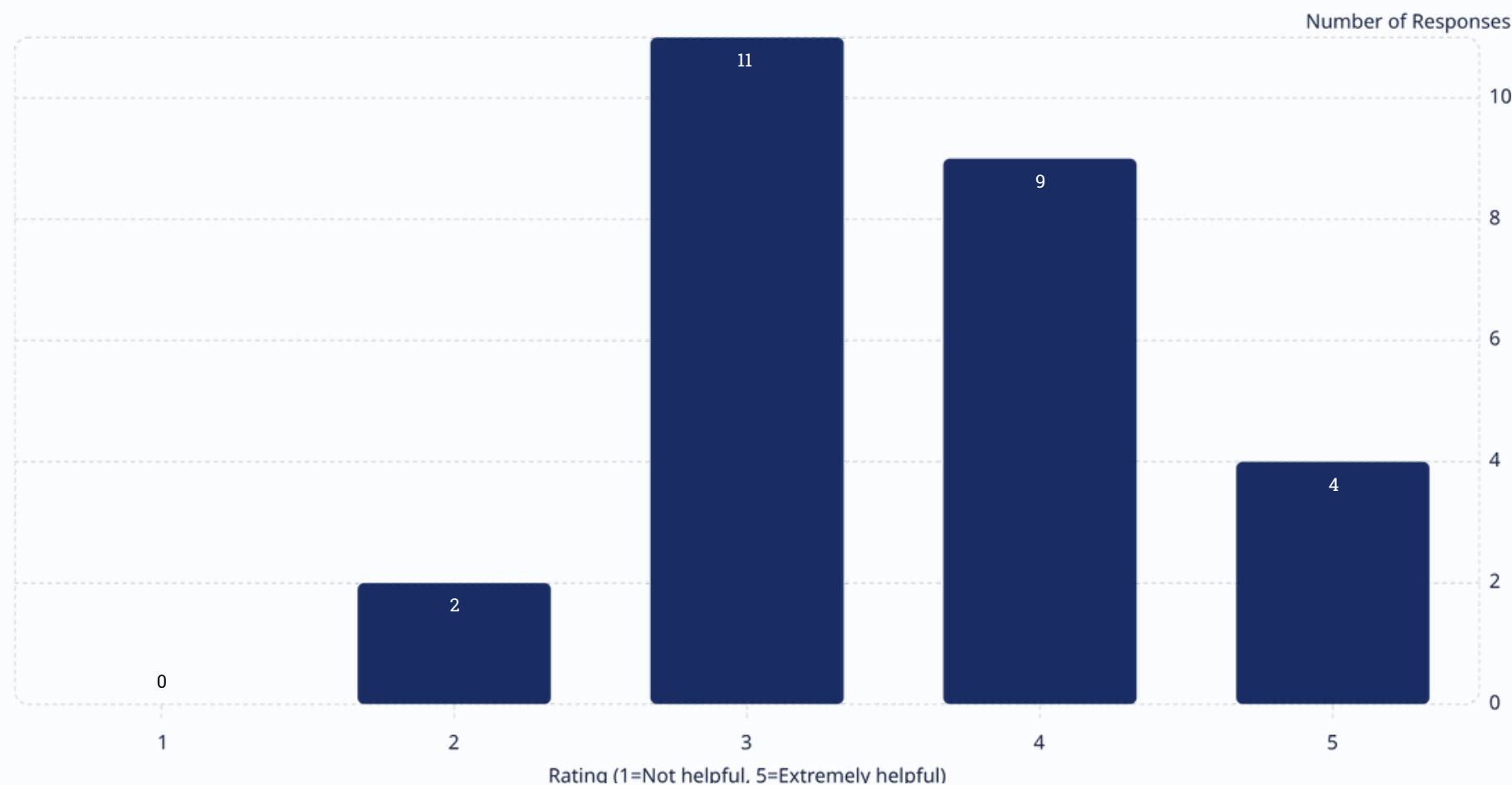
Key Finding

The largest group (**57.7%**) reports the app simply presents the correct answer and advances. About a third (30.8%) receive a varied follow-up question, and only **11.5%** are asked to apply the concept in a new context.

Encoding Specificity

Per Tulving and Thomson's Encoding Specificity Principle, memory transfers best when applied in varied contexts. The Encoding Variability Principle confirms that only 11.5% of learners receive such effective post-challenge practice — a significant gap.

How Helpful Is the Error Explanation?



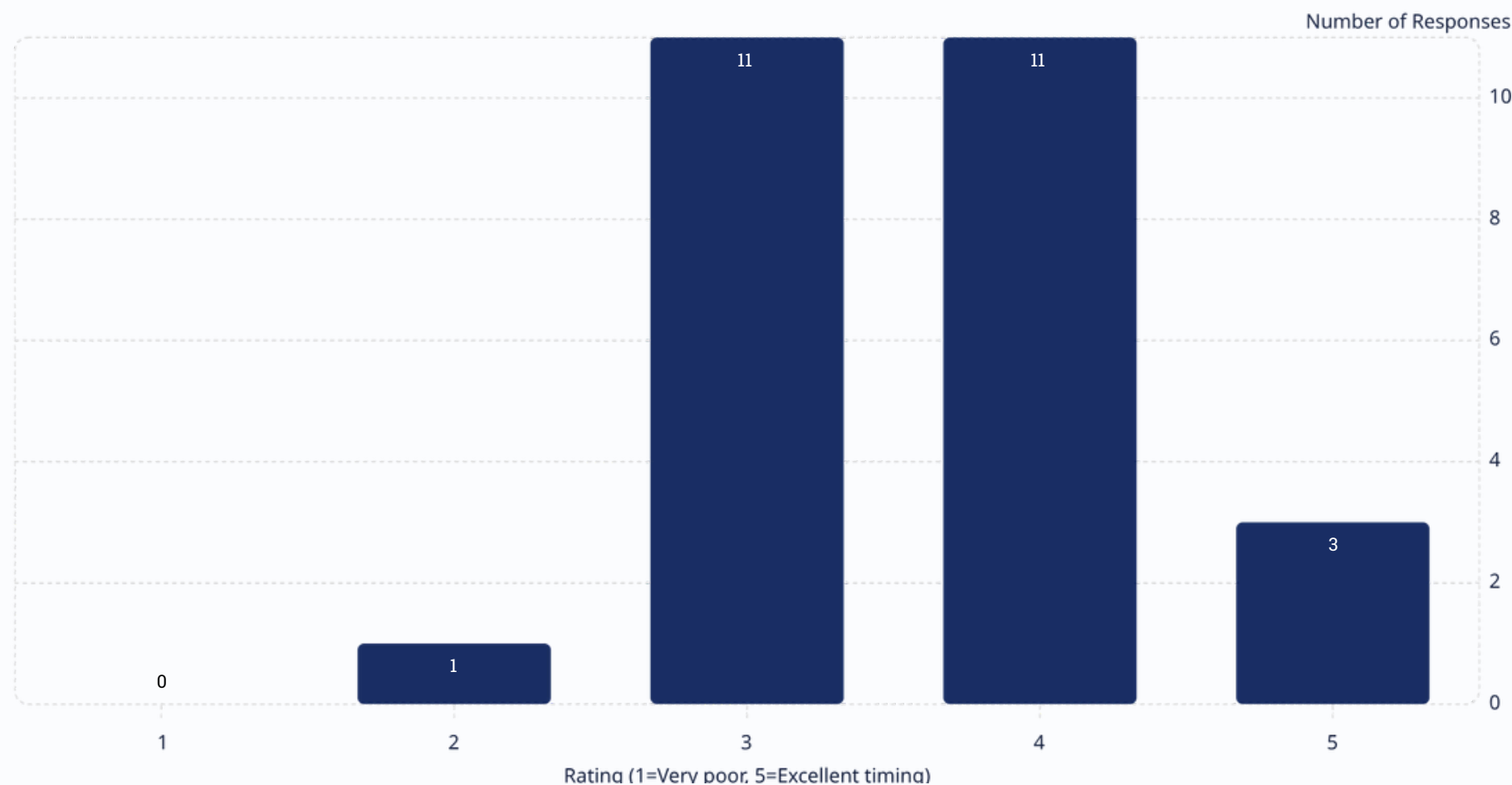
Key Finding

No respondent rated explanations as unhelpful. The distribution peaks at rating 3 (42.3%), with 34.6% at 4 and 15.4% at 5. **Mean = 3.58 / 5.**

Error-Driven Learning

Mistakes are highly effective for memory encoding — but only with *explanatory* feedback, not mere answer validation. The skew toward rating 3 suggests Coursera's error feedback is insufficiently concept-focused, leaving learners without full understanding of their mistakes.

How Effective Is the Timing of Feedback?



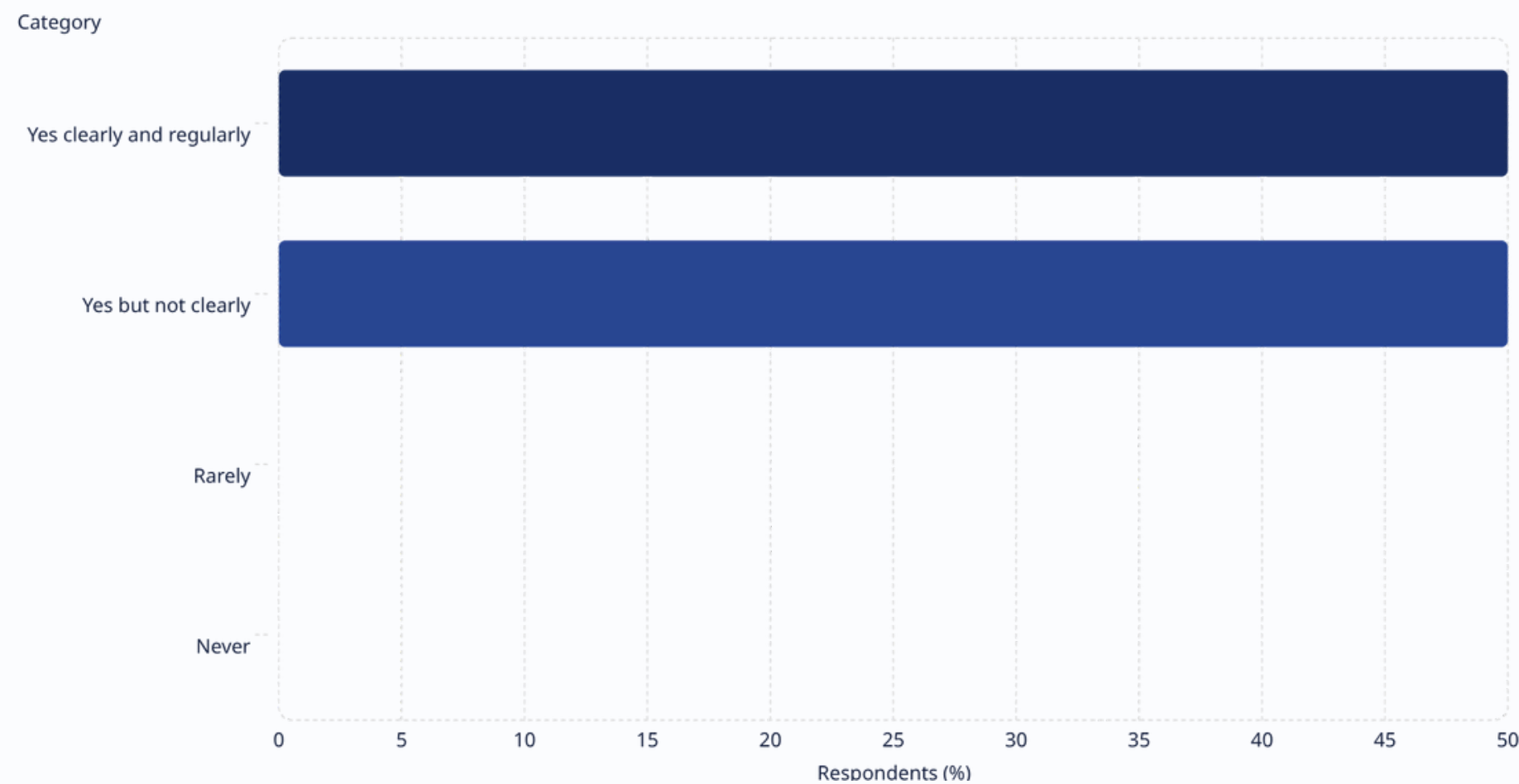
Key Finding

Ratings 3 and 4 are tied at 42.3% each, with 11.5% at 5. Only one respondent rated timing at 2; none at 1. **Mean = 3.62 / 5.**

Immediate vs. Delayed Feedback

Immediate feedback is more effective for error correction as the mental context remains fresh. However, the absence of very high ratings indicates delayed feedback, which is less effective, as the original mental context fades after a certain time.

Is Learning Progress Visible?

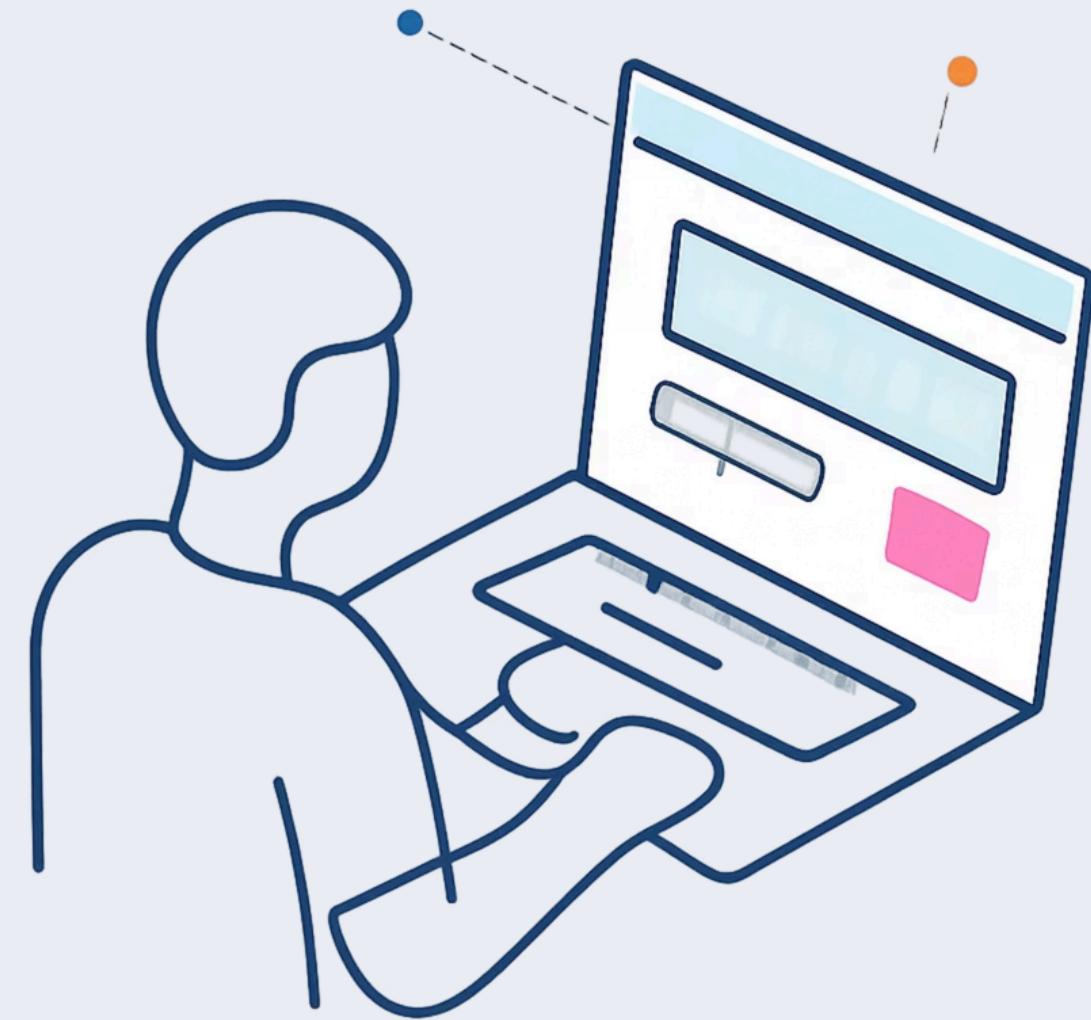


Key Finding

Responses split exactly **50-50** between "Yes, clearly and regularly" and "Yes, but not very clearly." No respondent selected Rarely or Never.

Metacognitive Calibration

Flavell (1979) identifies two purposes for progress visibility: motivational support and metacognitive calibration — accurate awareness of what learners know vs. still need to learn. As there isn't any votes for rarely or never, it shows that coursera is pretty good in terms of progress visibility. However the equal split in the graph shows that they can improve.



Neilsen's Principles

Q1: Visibility of System Status

"When you complete an activity or earn a reward, does the app make it immediately clear what just happened and what comes next?"

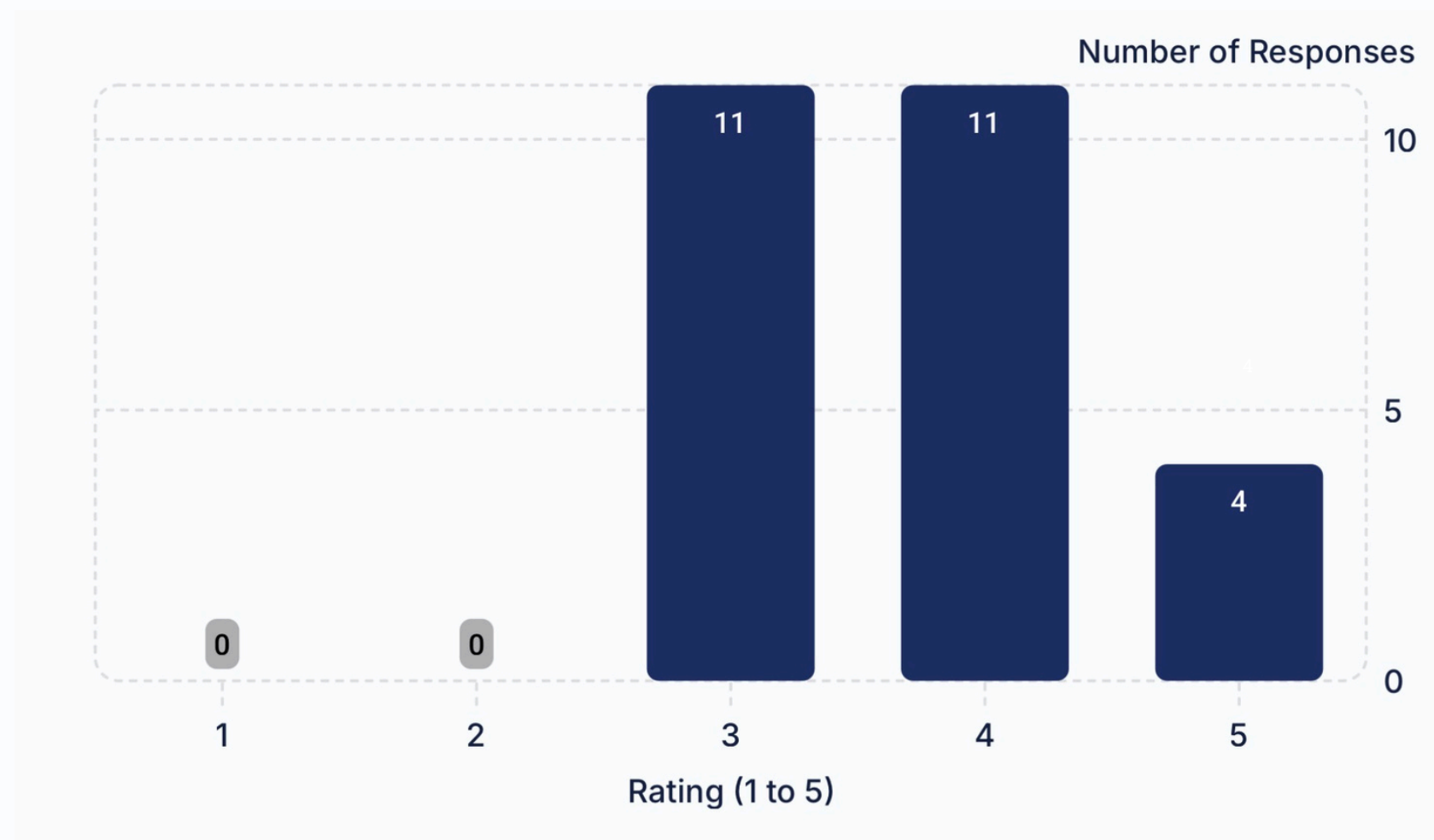


Key Finding

No negative ratings (1 or 2). 15/26 users gave positive scores (4 or 5); 11 gave neutral (3)

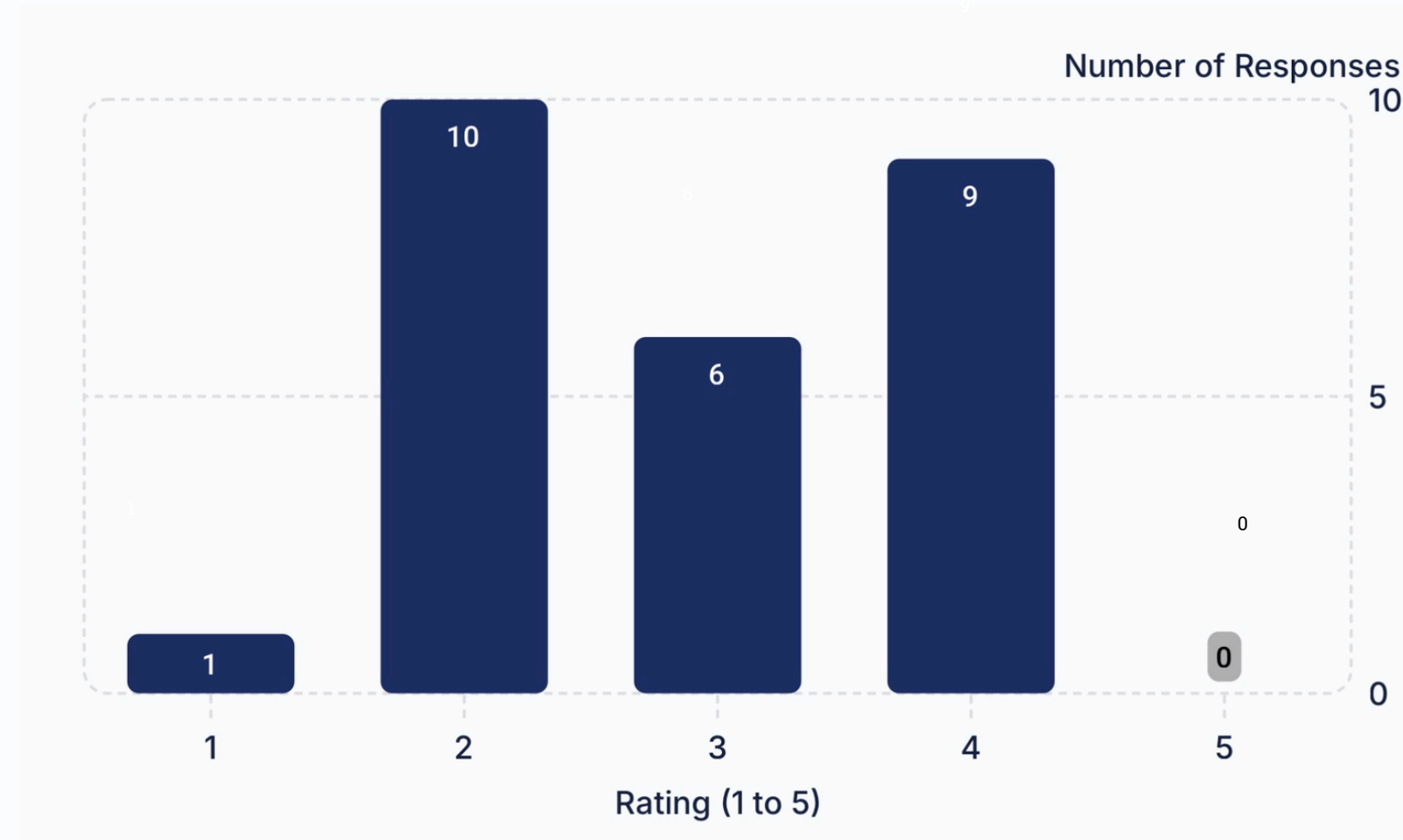
Behavioural Impact

- Users stop to verify progress instead of flowing to the next step
- No severe anxiety over lost progress, but hesitation occurs
- Absence of clear visuals (e.g., "Progress Saved!") reduces user momentum



Q2: Match Between System and the Real World

"Do the instructions, labels, and game concepts feel natural and easy to understand, or do they use jargon that makes you pause or guess?"



Key Finding

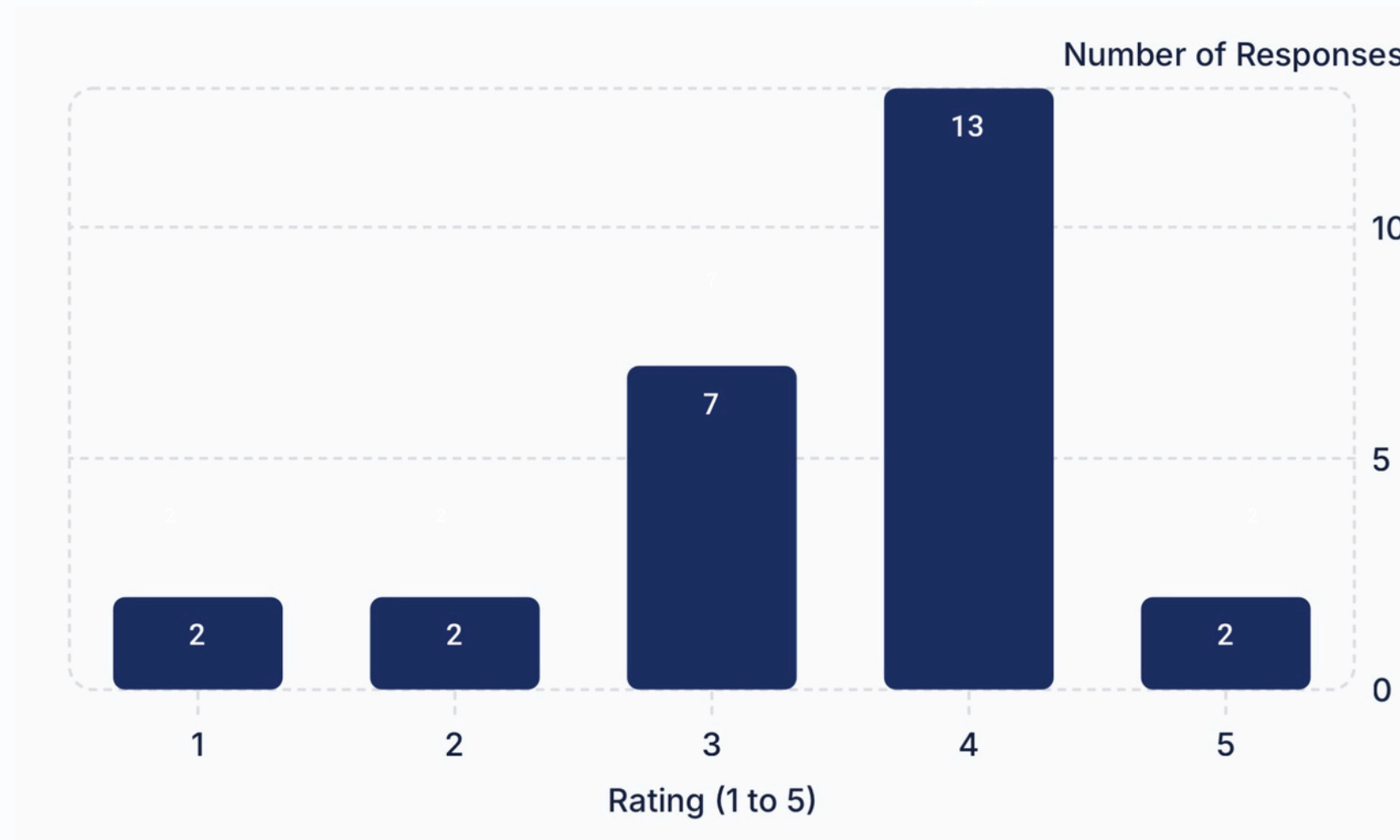
Highly polarized: **42% gave negative ratings** (11/26)
vs. 34% positive (9/26)

Behavioural Impact

- Unfamiliar jargon forces users to pause and guess what things mean
- Confusing labels lead to trial-and-error navigation
- Repeated unnatural terminology leads to fatigue and app abandonment

Q3: User Control and Freedom

"If you accidentally tapped the wrong answer or entered the wrong level, could you easily undo or go back — or did you feel stuck with a mistake?"



Key Finding

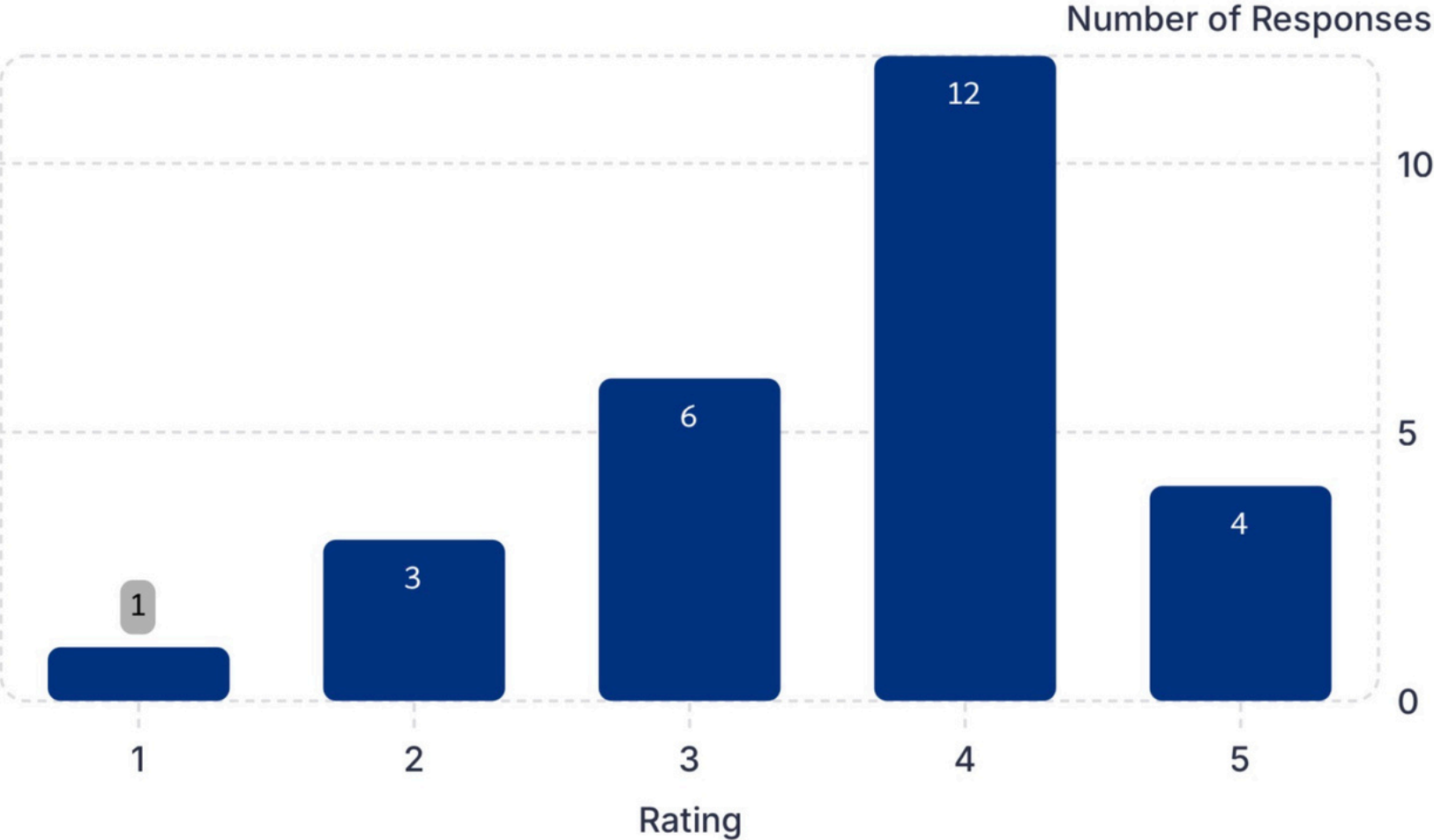
15 out of 26 users gave positive ratings; only 4 expressed frustration (1 or 2)

Behavioural Impact

- Users interact more confidently knowing they can recover from accidental taps
- Freedom encourages exploration without fear of permanent negative consequences
- A small frustrated subset suggests isolated screens may be missing back/undo buttons

Q4 – Consistency and Standards

"Have you ever been confused because a button, icon, or interaction worked differently in one part of the app than in another?"



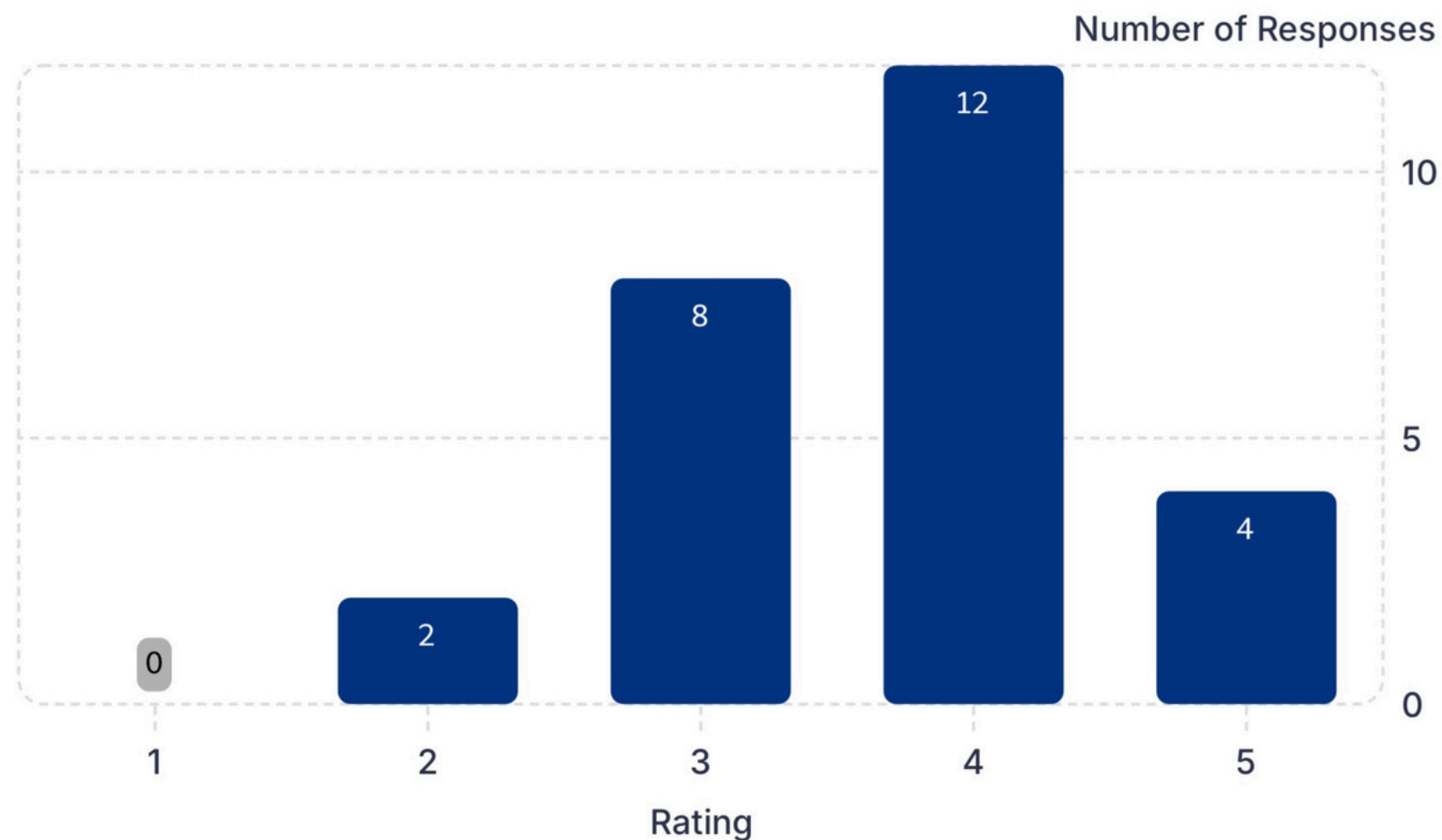
Key Findings

Mean score: **3.58/5**. Since this question is negatively framed, higher ratings indicate **poorer** adherence to consistency. The distribution skews toward ratings 4–5, revealing noticeable confusion from inconsistent interface behaviour.

⚠ Inconsistent signifiers weaken user trust and force users to relearn behaviour across sections, increasing cognitive effort and encouraging trial-and-error interactions.

Q5 – Error Prevention

"Does the app ever let you do something by accident — like submitting an incomplete answer or skipping a required step — without warning you first?"



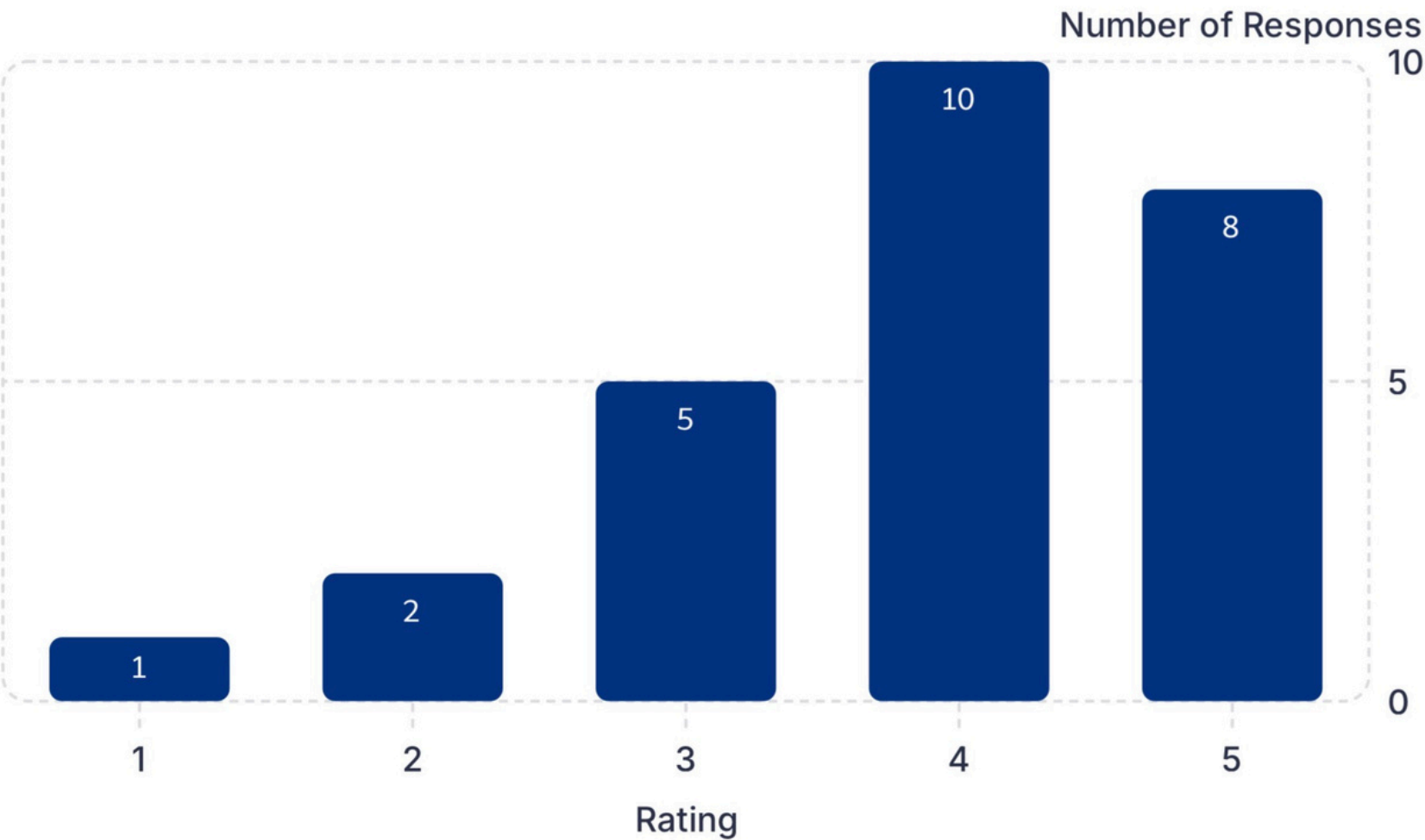
Key Findings

Mean score: **3.69/5**. A higher score here indicates **weaker** error prevention. The platform does not consistently stop users before avoidable mistakes such as incomplete submissions or unintended skips.

- ⊗ Users rely on self-checking rather than platform safeguards, increasing effort and frustration. Repeated accidental actions can make users overly cautious, slowing overall interaction.

Q6 – Recognition Rather than Recall

"When you return to the app after a break, can you quickly pick up where you left off, or do you have to remember a lot of context on your own?"



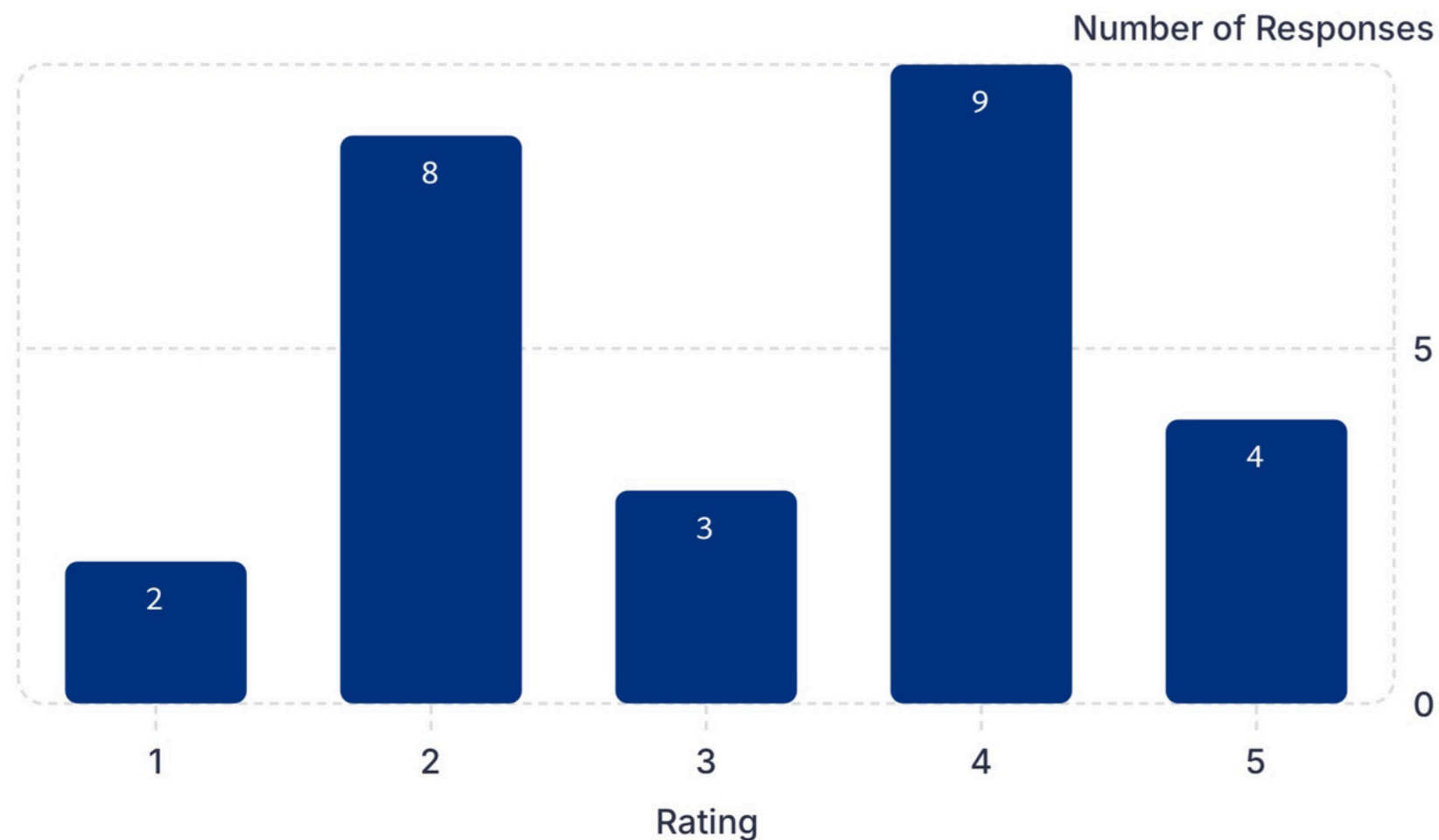
Key Findings

Mean score: **3.85/5** — the one of the strongest result. Higher ratings here represent **better** usability. Progress indicators, structured modules, and navigation support are functioning reasonably well.

- ✔ Users re-enter the learning flow with low memory burden. Visible structure supports confidence and continuity, improving persistence by reducing the mental effort to restart a session.

Q7 – Flexibility and Efficiency of Use

"Are there shortcuts, customization options, or ways to skip content you already know — or does the app force everyone through the same path?"



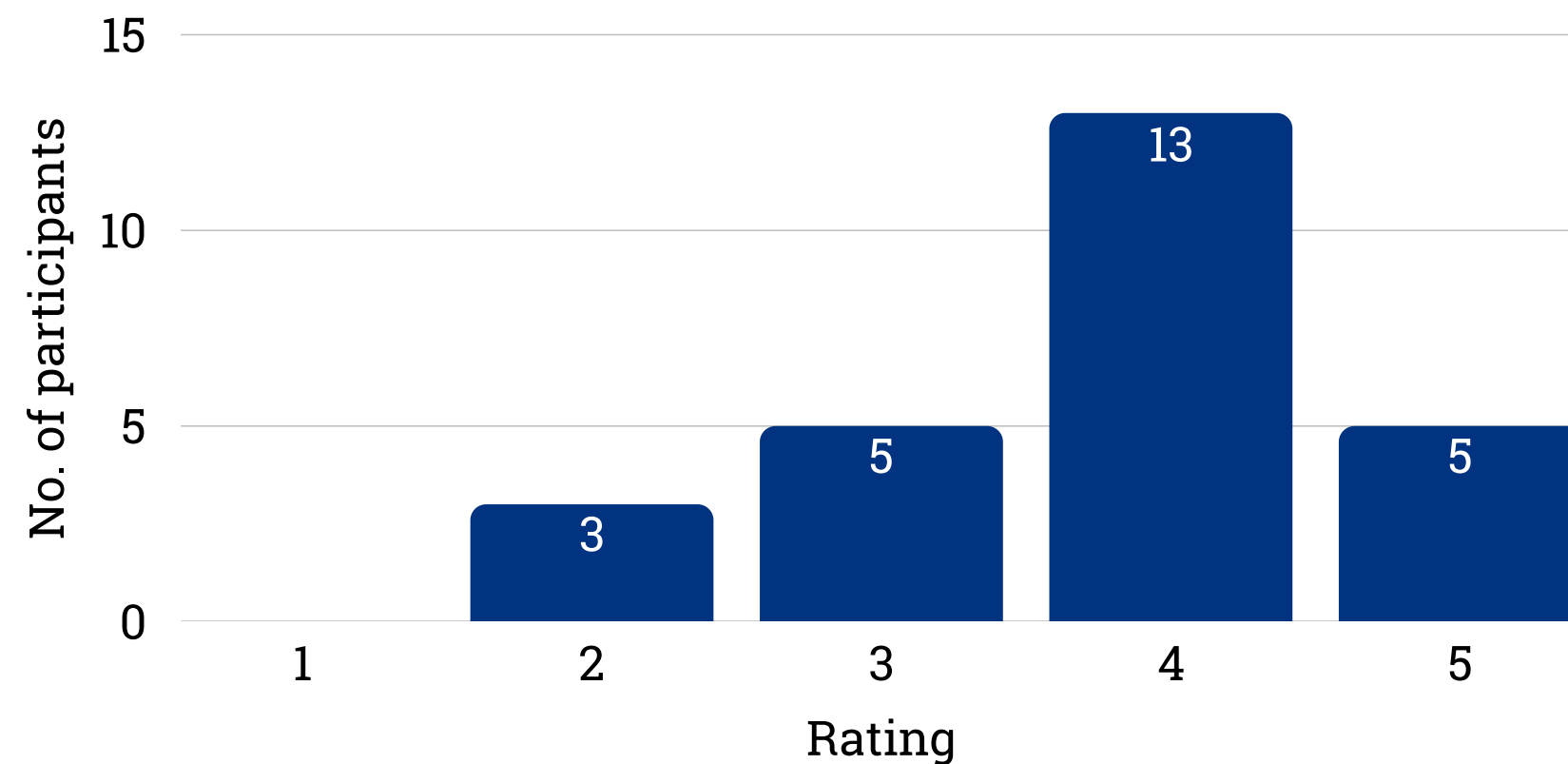
Key Findings

Mean score: **3.19/5** — the lowest among Q4–Q7. Responses are spread across the scale, indicating a mixed experience. The platform supports baseline usability but lacks strong personalization or efficient shortcuts for varied skill levels.

⚠ Experienced users feel slowed by fixed paths. A one-size-fits-all approach does not align with varied learning preferences or expertise levels.

Q8 – Aesthetic and Minimalist Design

“Have you ever felt overwhelmed by too many elements on the screen at once notifications, scores, timers, instructions competing for your attention?”



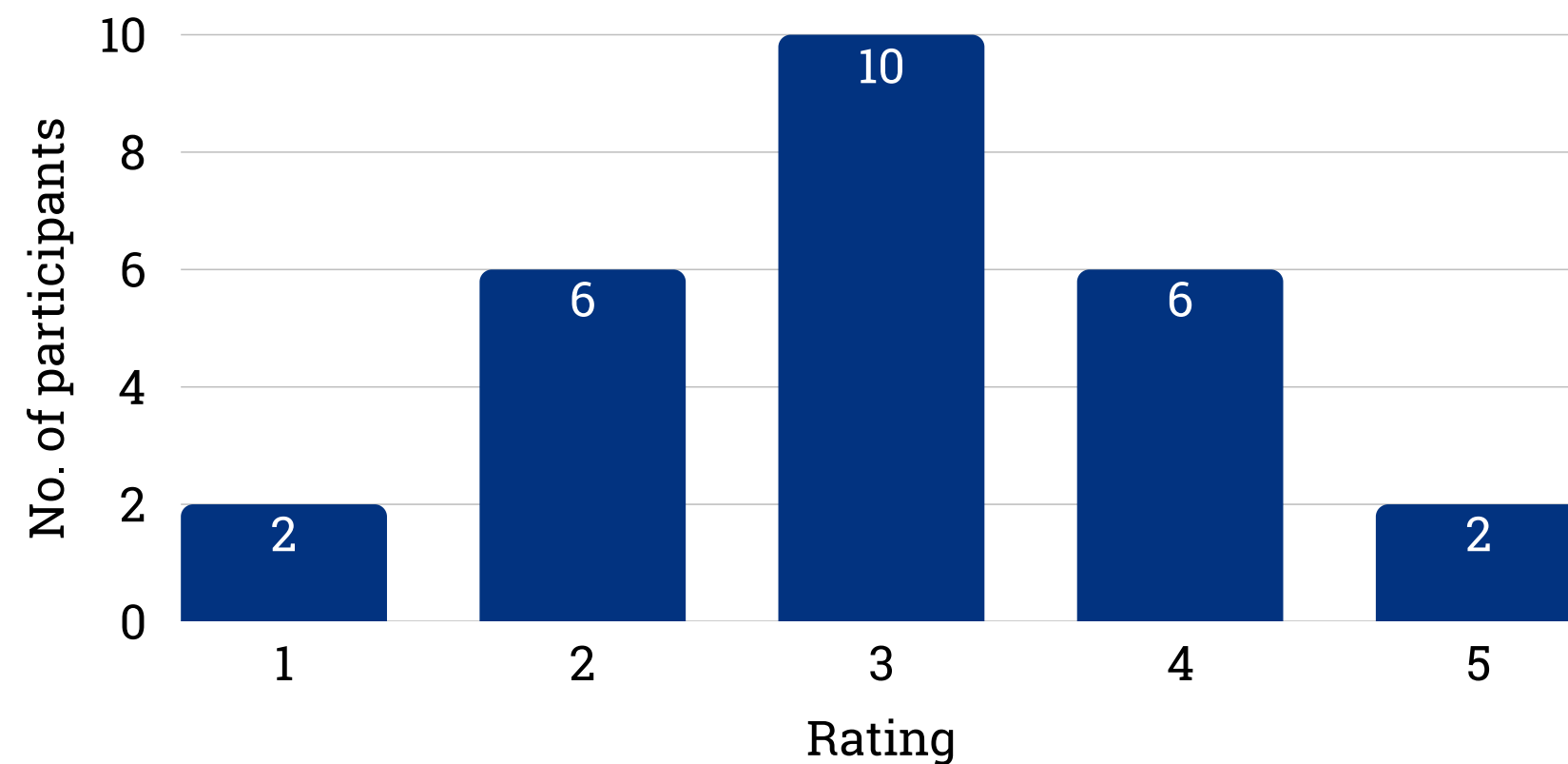
Key Findings

Mean score: ~3.77/5 — skewed towards higher ratings, indicating **frequent visual overload**. 50% of users rated 4, with additional responses at 5 (19.2%), suggesting that clutter is a common issue rather than an exception.

- ⊗ Users often feel overwhelmed due to multiple competing UI elements. Lack of visual hierarchy increases cognitive load and reduces focus on core tasks.

Q9 – Help Users Recognize, Diagnose & Recover from Errors

“When something goes wrong (a wrong answer, a failed submission, a loading error), does the app explain what happened and how to fix it in plain language?”



Key Findings

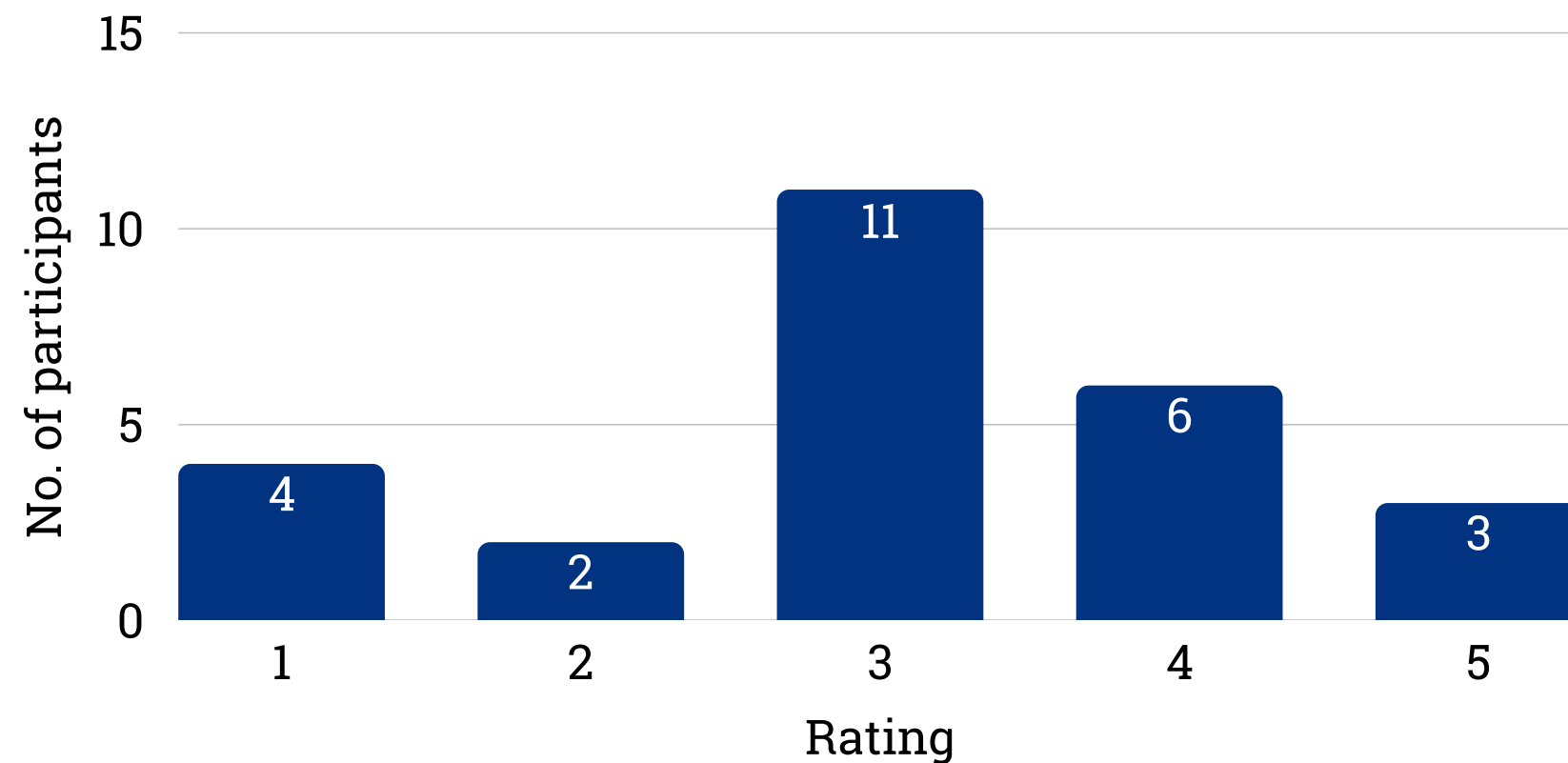
Mean score: ~3.00/5 — indicating a neutral and inconsistent experience. Responses are evenly distributed around the midpoint, with 38.5% rating 3 and equal proportions (23.1%) at 2 and 4, reflecting **variability in error communication**.



Users partially understand errors but lack clear recovery guidance. Inconsistent messaging leads to trial-and-error behaviour instead of confident correction.

Q10 – Help and Documentation

“If you were confused about how a feature works, was it easy to find guidance within the app or did you have to search outside of it (YouTube, forums, asking someone else)?”

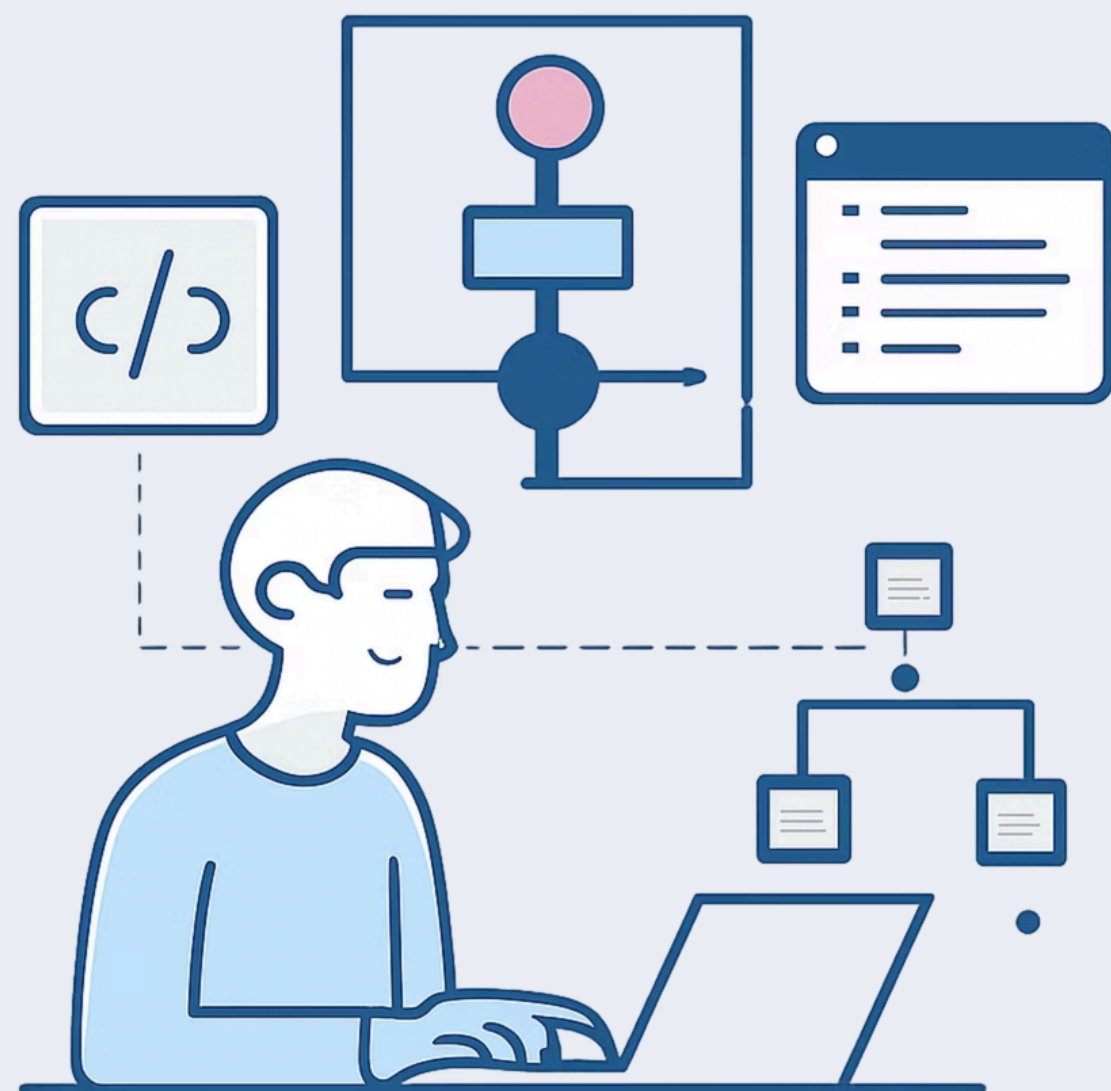


Key Findings

Mean score: ~3.08/5 — indicating **moderate but inefficient access to help**. 42.3% of users rated 3, while 23.1% rated low (1–2), showing that a significant number of users struggle to find in-app guidance.

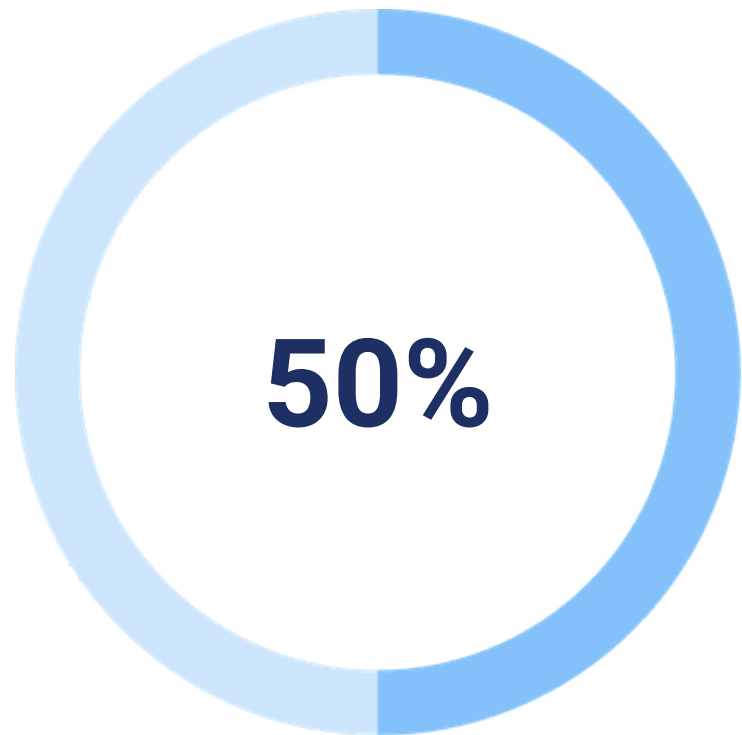


Users rely on external resources due to poor discoverability of help features. Guidance exists but is not easily accessible within the app.



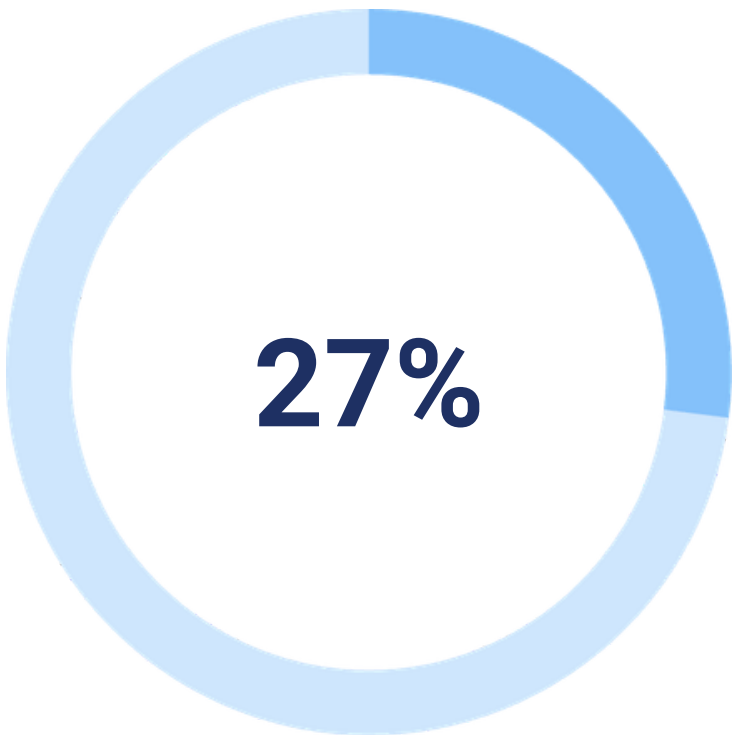
Real World Problems

Q1: What do users do when they face a challenge on the app?



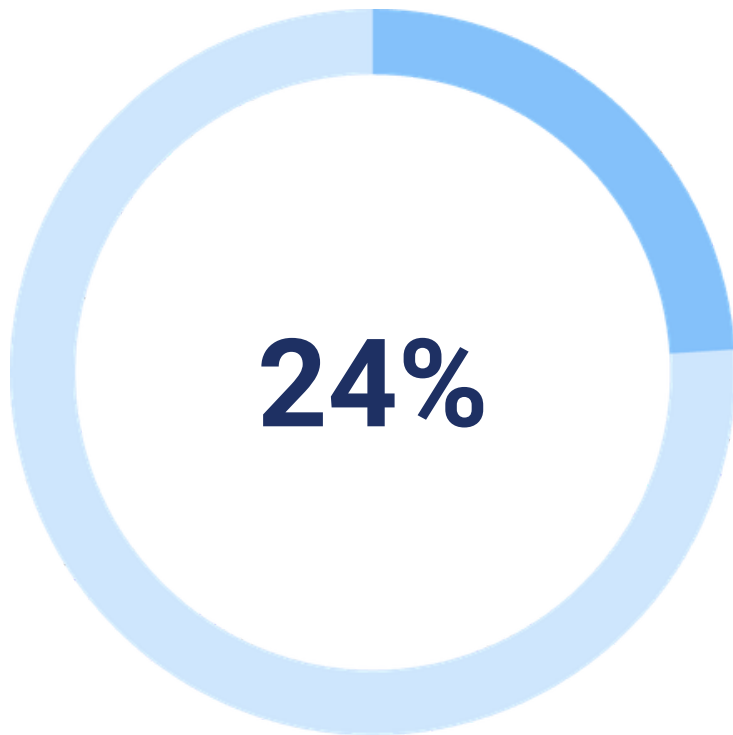
Persist Through

Push through challenges with high frustration tolerance



Disengage Completely

Stop using the app entirely when stuck



Skip Challenges

Bypass difficult content to maintain progress

Persistence Under Duress

Half of users force themselves through challenges, raising questions about difficulty calibration. When users push through rather than enjoy the challenge, it indicates poor flow state design and potential frustration.

Risk of Dropout

27% disengagement signals need for better scaffolding mechanisms like hints, checkpoints, or adaptive difficulty to prevent permanent abandonment.

Q2: When and why did the users start feeling rewards were meaningless?



Lack of Real-World Value

Rewards carry no transferable weight outside the platform. Users recognised points as "just numbers" with no utility on LinkedIn or resumes, causing instrumental value collapse.



Trivial Achievement

Badges awarded for minimal effort fail to trigger satisfaction from genuine accomplishment. Easy acquisition signals low competence rather than real achievement.



Timeframe Matters

Some users felt rewards meaningless from the start. Others experienced disillusionment after 2-3 weeks as repetitive nature became apparent.

Q3: Did the rewards distract the user from actual learning?

Game elements designed to motivate users instead became the primary focus, displacing genuine learning as the central goal for many respondents.

Incentive Misalignment

Users admitted to rushing through content, guessing answers rapidly, or seeking external help to maximise points. The measure became the target, creating counterproductive loops where gamification incentivises behaviours that hollow out learning.

Completion Over Comprehension

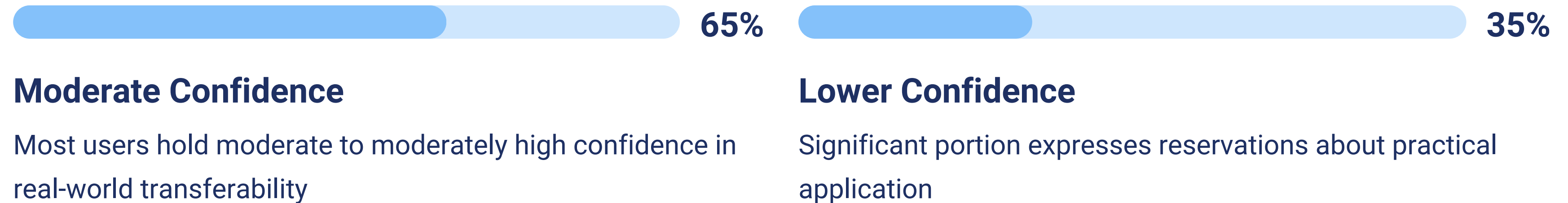
Several users rushed through material to obtain completion certificates before access expired or maintain unbroken streaks. External pressures caused loss of sight of inherent value, prioritising consistency over quality of engagement.

False Confidence from Guessing

When users guess correctly and receive points, the system reinforces misconceptions, treating lucky guesses as equivalent to genuine knowledge. Timed quizzes introduce cognitive load that diverts mental resources from thoughtful engagement to rapid responses.

Q4: User's confidence in the real-life transfer of the learnings from the app

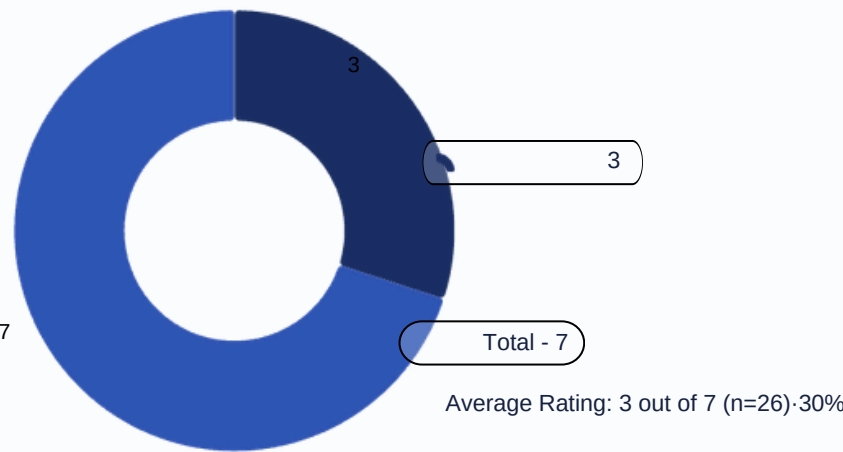
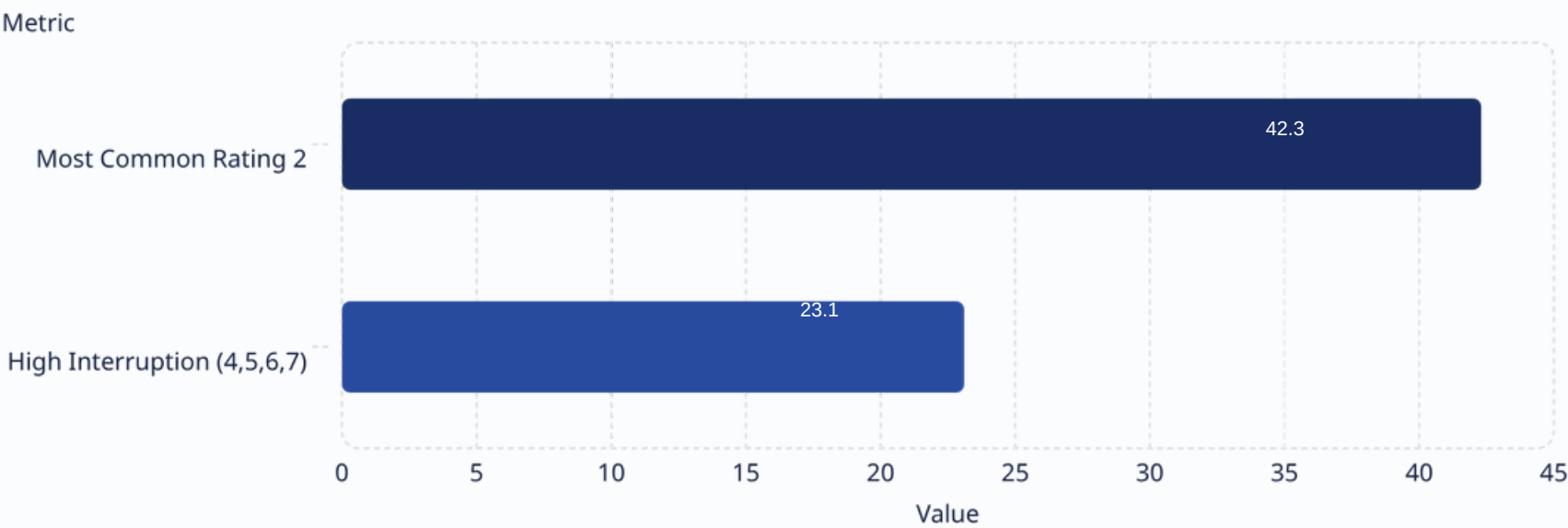
Users were asked to rate confidence that app practice transfers to real-life situations or exams. Distribution shows moderate to moderately high confidence, but lack of extremities reveals uncertainty.



Skewed distribution toward ratings 4-6 indicates most users hold moderate confidence, but absence of extreme certainty suggests need for stronger evidence of practical value.

Q5: Access & Equity - Mid-Session Interruptions

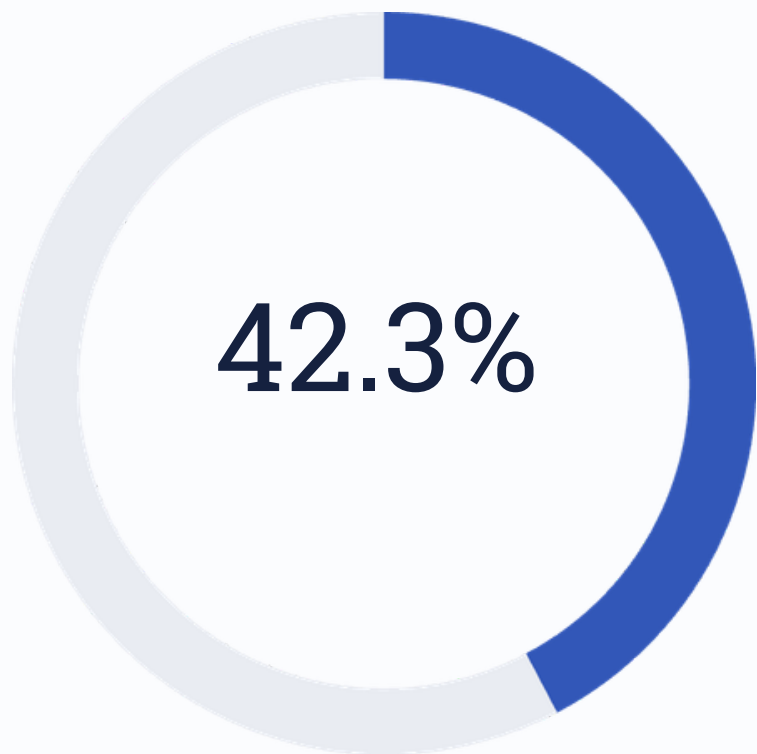
How often did you have to stop using the app mid-session because of weak internet, low battery, or shared device?



Most learners reported **low-to-moderate** interruptions, but **not everyone** has the same experience. A smaller group faces **frequent breaks**, which hurts focus and makes **learning slower**. So, the same app design works differently for different users because of device/internet conditions.
Main inference: Access issues create unequal learning outcomes even when motivation is similar.

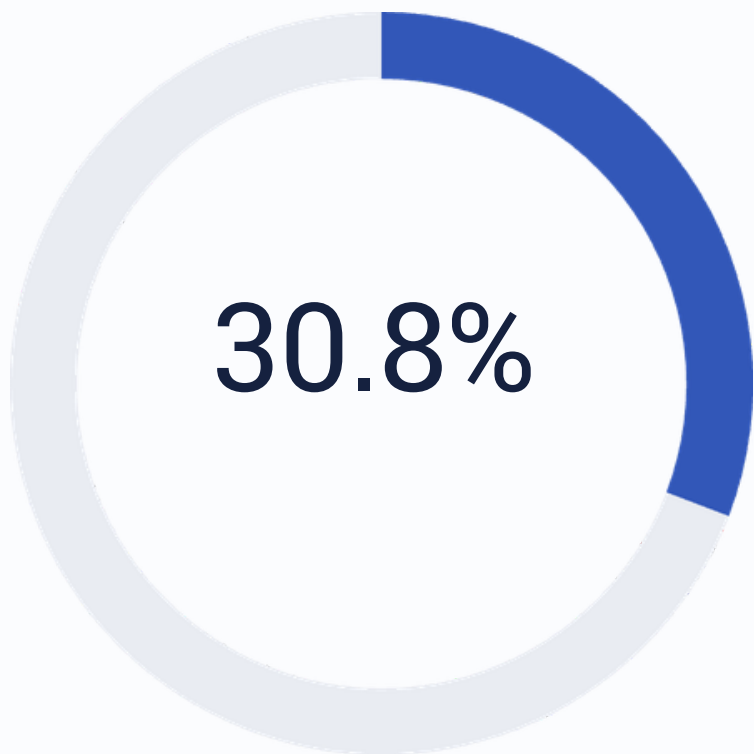
Q6: Access & Equity - Learner-Content Mismatch

Are there features or content that feel designed for a different kind of learner in terms of language, cultural references, or learning pace?



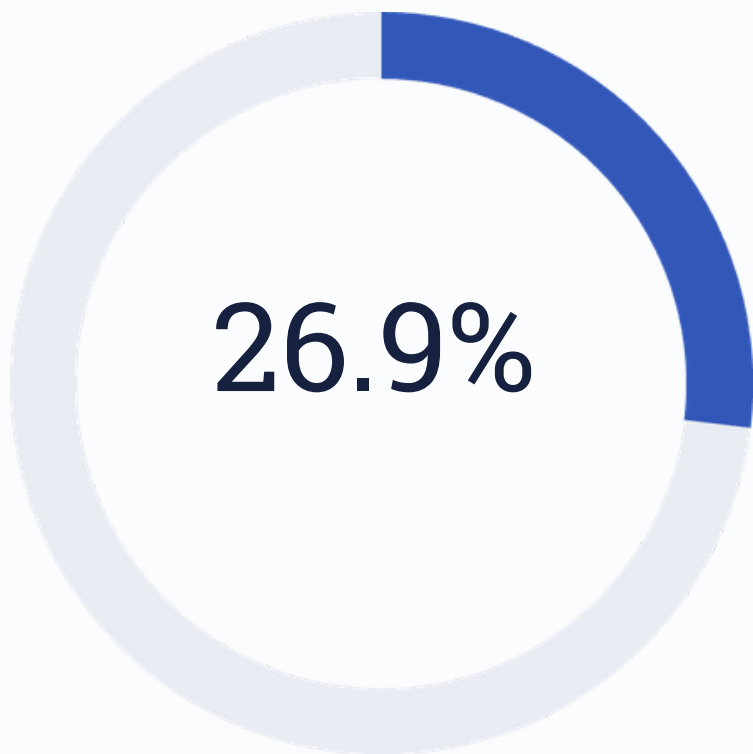
Maybe

Intermittent mismatch



Yes

Perceived mismatch



No

Good fit

Many users said "Maybe", which means mismatch is present but not always obvious. Some content **seems to fit**, but certain examples/language/pacing do not **suit all learners**. This **extra adjustment** effort can reduce confidence and make learning **feel harder** than it should.

Main inference: The app needs more flexible content style to support diverse learners.

Q7: Social Dynamics - Leaderboard Discouragement

Have leaderboards or comparisons with other users ever made you feel discouraged rather than motivated? What did you do?



Not Discouraged

14 users (53.8%)

Little/no leaderboard exposure or neutral experience



Motivated

4 users (15.4%)

Comparison energizing and performance-enhancing



Discouraged

6 users (23.1%)

Temporary withdrawal or rushing behavior



Mixed

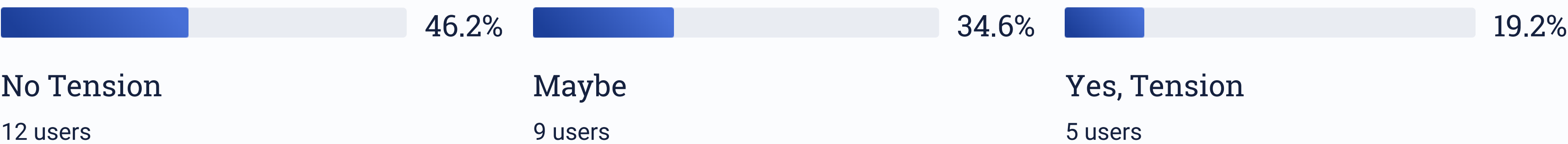
2 users (7.7%)

Self-regulatory buffering strategies

More than half were not discouraged, so comparison is not harmful for everyone. But a **meaningful group** felt discouraged and sometimes disengaged when comparison pressure increased. Some users are motivated by rankings, while others prefer focusing on **personal progress**.
Main inference: Competitive features should be optional/personalized, not one-size-fits-all.

Q8: Social Dynamics - Competitive Tension

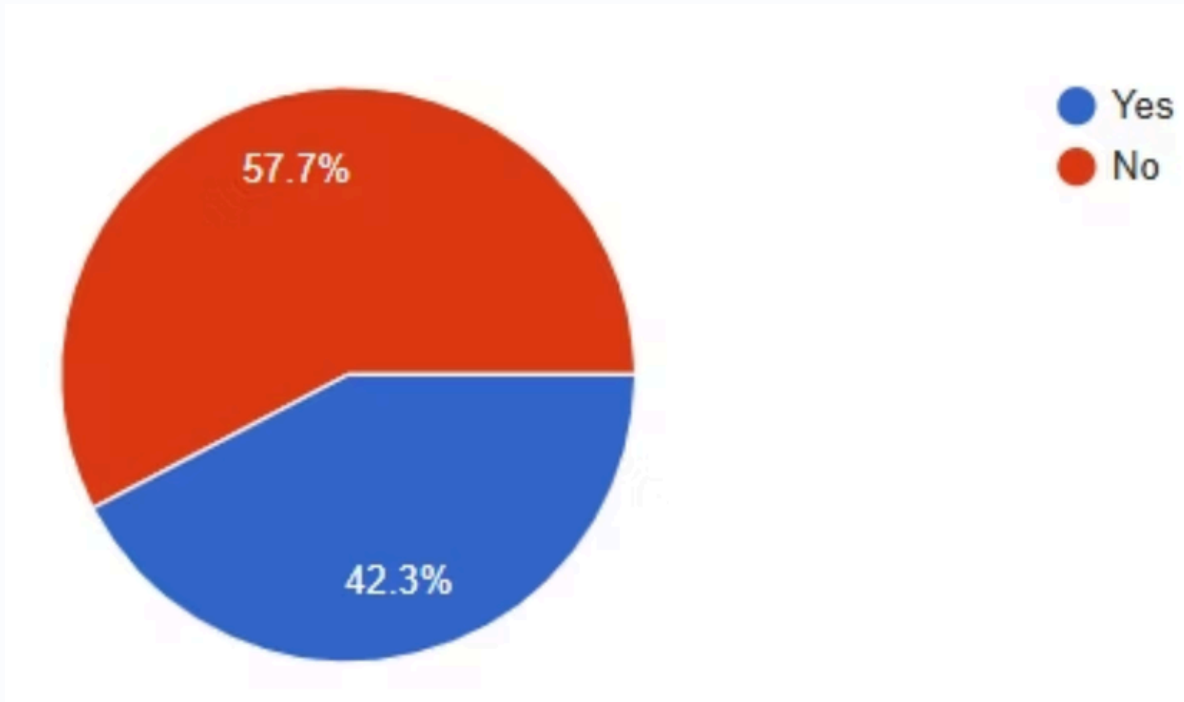
If using the app in a classroom or group setting, does the competitive structure cause tension between students?



Most users did not report regular tension, but many still said "Maybe" or "Yes." This means **conflict is not constant**, but appears in certain group contexts. Competitive structures can reduce collaboration when students feel judged or compared too strongly.

Main inference: Balance competition with **teamwork** and **mastery-focused** feedback to avoid peer tension.

Q9: Do Users Trust the App's Progress Tracking?



What Users Experienced

- **User-System Disconnect:** Answers were marked wrong even when users believed they were correct, creating a user-system disconnect
- **Increased Cognitive Load:** Progress was not saved between sessions, forcing users to double-check their status and increasing cognitive load
- **Motivation & Abandonment:** Feedback often did not align with actual performance, which reduced trust, lowered motivation, and could lead to abandonment

Recommendations

Diagnostic Review

The team can conduct an audit to determine if these reports stem from technical bugs or ambiguous UI design that misleads user input.

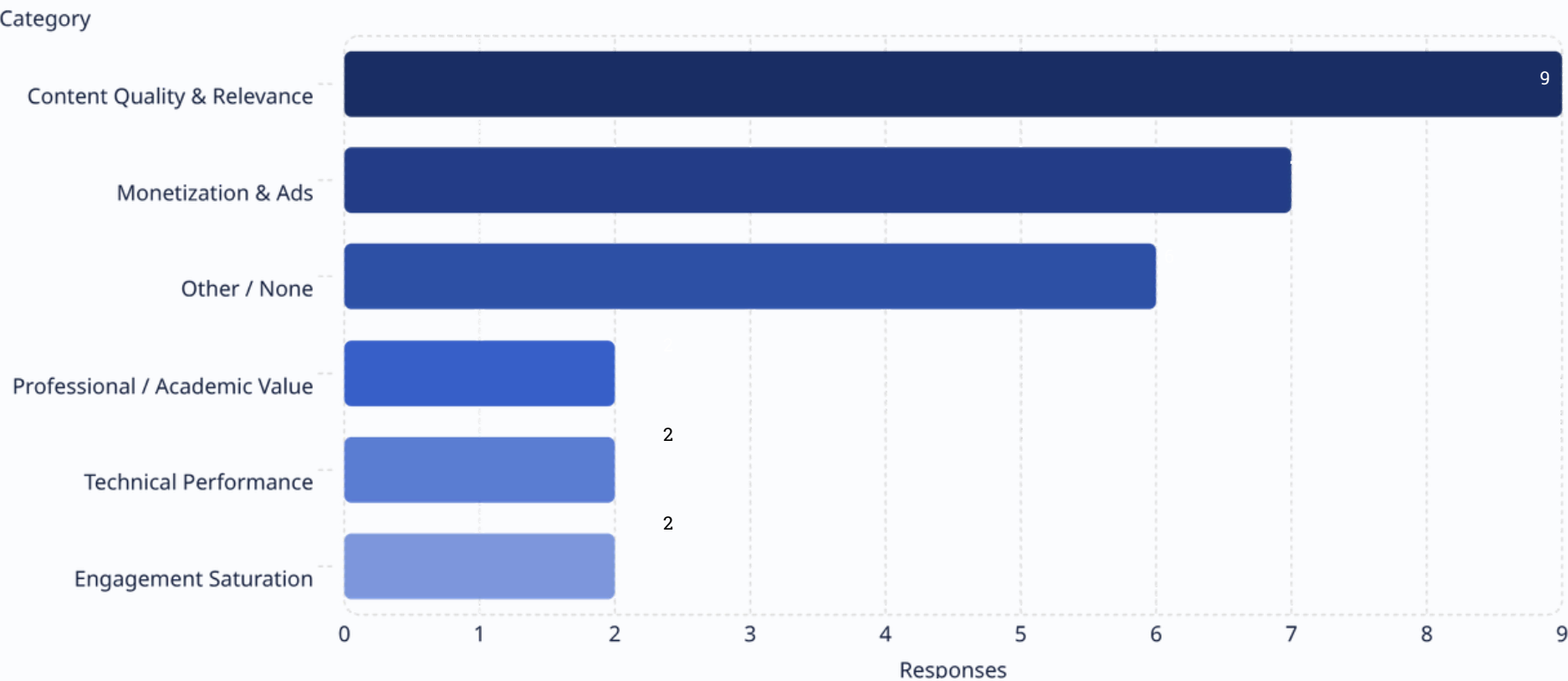
Enhanced Feedback Mechanisms

The team can then implement more granular explanations for incorrect answers to help users reconcile their reasoning with the system's logic.

Reporting Channels

They can provide a direct, low-friction mechanism for users to flag potential errors, transforming technical frustration into actionable insights for the development team.

Q10: What Would Make You Stop Using the App Permanently?



Top Abandonment Drivers

- **Content:** Outdated material, poor quality, lack of real-world applicability, insufficient skill development
- **Monetization:** Excessive charges, paywalls, end of course scholarships, payment for every course
- **Value:** Certifications losing market standing or better alternatives emerging
- **Engagement:** Excessive promotion of paid content, unnecessary ads for upgrades and gamification elements
- **Technical/Performance Issues:** App being slow or laggy

Reasons

Monetization as Cognitive Interrupters

When paywalls, charges, or interruptions appear too often, they break the learning flow and create frustration.

Gamification's Fading Effect

What once felt motivating can become predictable or hollow, reducing engagement instead of sustaining it.

Value Collapse Against Free Alternatives

If certifications lose credibility or better free options emerge, users may stop seeing enough reason to stay.

Q11: Net Promoter Score

42.3%

Detractors
Rated 0–6

42.3%

Passives
Rated 7–8

15.4%

Promoters
Rated 9–10

-26.92%

NPS
(n=26)

Interpreting the NPS



Functional Utility Only

Users see the app as useful for specific tasks, but not compelling enough to actively recommend by word-of-mouth



Bimodal Sentiment

Despite a plurality of users rating the app an 8 (30.8%), a significant percentage (42.3%) fall into the "Detractor" category



Low Advocacy Readiness

The current experience does not create enough delight or trust to convert users into promoters.



Key Takeaways

Key Takeaways

Our comprehensive analysis of Coursera's platform revealed key findings across memory science, usability, and real-world user experience.

Memory & Learning Science

Initial content encoding is strong, leveraging clear verbal and visual explanations. However, the platform significantly lacks effective mechanisms for long-term retention, with minimal spaced repetition and passive review methods, contributing to low confidence in longer retention.

Nielsen's Usability Heuristics

The platform performs well in "Recognition Rather than Recall" and "User Control." Key weaknesses include visual clutter (Aesthetic Design), inconsistent interface behavior, insufficient error prevention, and gamified terminology that often misaligns with user mental models.

Real-World Problems

Gamification's instrumental value collapses, leading to disengagement and shallow learning. Significant concerns around access, cultural alignment, and the polarizing effect of social comparison were noted. Users also expressed doubts about progress tracking accuracy, reflected in a Net Promoter Score of -26.92.



Our Learnings

Learning from the Project



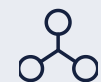
The Gap Between Intent & Implementation

Even reputable platforms widely praised for content can systematically violate established principles of learning science and usability. Good content does not guarantee good cognitive design.



Diverse User Needs & Contexts

User behavior is heterogeneous. Features like gamification or leaderboards affect individuals differently. Effective platform design requires adaptive personalization over uniform experiences.



The Power of Interdisciplinary Analysis

No single framework is sufficient. Rich insights emerge from triangulating across memory science, heuristic evaluation, and motivation theory for a holistic product evaluation.



Cognitive Science as Actionable Design Criteria

Cognitive science principles are not merely academic; they are measurable design criteria. Deeply integrating these principles from the ground up leads to genuinely better learning outcomes.



Thank you!